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**Human-Centered Leadership and Communication in Hybrid
Project Teams: An Exploratory Study of Team Dynamics and
Collaboration**

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GENERAL INTRODUCTION

Project management has evolved considerably in recent years as organizations operate in increasingly complex, dynamic, and uncertain environments. Beyond the traditional focus on planning, scheduling, and technical execution, contemporary project management now places greater emphasis on human-centered factors such as leadership practices, communication processes, and team dynamics. These factors play a critical role in shaping collaboration, engagement, and performance within project teams.

The widespread adoption of hybrid work arrangements has further transformed the way project teams' function. Project team members often collaborate across physical and virtual spaces, relying heavily on digital communication tools and asynchronous interactions. While hybrid work models offer flexibility and access to diverse talent, they also introduce new managerial challenges related to coordination, trust-building, shared understanding, and conflict management. In such environments, the effectiveness of leadership and communication becomes central to ensuring that project teams remain aligned, motivated, and collaborative.

Despite the availability of advanced project management tools and standardized methodologies, many projects continue to face difficulties such as miscommunication, reduced team cohesion, unclear roles, and collaboration breakdowns. These challenges are rarely caused by technical limitations alone but are often linked to how project managers lead teams, communicate with stakeholders, and manage interpersonal dynamics within hybrid settings. As a result, project success increasingly depends on the ability of project managers to adopt human-centered approaches that address the social and relational dimensions of teamwork.

Academic literature highlights the growing importance of soft skills, emotional intelligence, inclusive leadership, and effective communication in project management. Prior studies suggest that human-centered leadership styles can positively influence team motivation, trust, and performance, particularly in diverse and distributed teams. However, while these themes are widely discussed, there remains a gap in empirical research examining how leadership and communication practices concretely shape team dynamics and collaboration in hybrid project teams operating within today's evolving work environments.

This gap presents a significant managerial challenge. Project managers are expected to lead hybrid teams effectively, foster collaboration, and manage conflicts, yet they often lack clear, evidence-based guidance on which leadership behaviours and communication strategies are most effective in practice. Without a deeper understanding of how human-centered leadership and communication influence team dynamics, organizations risk reduced project performance, lower team engagement, and weakened collaboration.

In response to this challenge, this thesis adopts an exploratory approach to examine human-centered leadership and communication in hybrid project teams. By focusing on team dynamics and collaboration, the study aims to provide insights into how project managers can better navigate the human complexities of hybrid work environments. Through qualitative investigation, this research seeks to contribute to both academic understanding and managerial practice by identifying patterns, challenges, and effective practices that support collaboration and team effectiveness in contemporary project management contexts.

Research Problem

How can project managers adopt human-centred leadership and communication practices to enhance team dynamics and collaboration in hybrid project teams?

Research Questions

To address the research problem, this study is guided by the following research questions:

- * **RQ1.** How do project managers describe and apply human-centred leadership practices within hybrid project teams?
- * **RQ2.** How are communication processes structured and experienced in hybrid project teams, and what challenges do they present for effective collaboration?
- * **RQ3.** How do leadership and communication practices influence team dynamics, including trust, motivation, and conflict management, in hybrid project environments?
- * **RQ4.** What practices are perceived by project managers as most effective in enhancing collaboration and cohesion within hybrid project teams?

PART 1 – LITERATURE REVIEW

Introduction

This part of the thesis aims to establish the theoretical foundations necessary to address the research problem by reviewing and synthesizing existing scientific literature related to human-centered leadership, communication in project teams, team dynamics and collaboration, and hybrid project environments. The purpose of this literature review is not merely to summarize prior studies but to critically analyze key concepts, theoretical perspectives, and empirical findings that inform the understanding of how leadership and communication influence collaboration within contemporary project teams.

Project management literature has progressively shifted from a predominantly technical and process-oriented focus toward a more human-centered perspective. While traditional project management emphasized planning tools, control mechanisms, and performance metrics, recent research highlights the critical role of leadership behaviors, communication practices, and interpersonal relationships in determining project success, particularly in complex and uncertain environments (Müller et al., 2019) (Dimitroff et al., 2005). This shift is especially relevant in the context of hybrid project teams, where collaboration relies heavily on social interaction, trust, and effective communication across physical and virtual boundaries.

Accordingly, this literature review is structured around four main thematic areas. First, it examines human-centered leadership in project management, focusing on the evolution of leadership approaches and the growing importance of people-oriented leadership styles. Second, it explores communication in hybrid project teams, emphasizing the role of communication as a coordination and alignment mechanism. Third, it reviews literature on team dynamics and collaboration in project environments, including trust, motivation, and conflict management. Finally, it situates these themes within the broader context of hybrid project teams as a contemporary work arrangement.

By synthesizing insights across these streams of literature, this part identifies key theoretical foundations as well as gaps in current research, particularly regarding the practical application of

human-centered leadership and communication in hybrid project settings. These gaps provide the justification for the empirical study presented in Part 2 of the thesis.

CHAPTER 1

1 REVIEW OF SCIENTIFIC LITERATURE

1.1 Human-Centered Leadership in Project Management

1.1.1 Evolution of Leadership in Project Management

Leadership has long been recognized as a critical factor in project success. Early project management research primarily viewed leadership through a task-oriented and control-based lens, emphasizing planning accuracy, monitoring, and adherence to predefined schedules and budgets (Kerzner, 2025). In this traditional perspective, project managers were expected to act as coordinators and controllers, ensuring that tasks were executed efficiently and deviations were corrected promptly.

However, as projects have become increasingly complex, uncertain, and knowledge-intensive, scholars have questioned the adequacy of purely task-focused leadership approaches. Contemporary projects often involve diverse stakeholders, cross-functional teams, and high levels of interdependence, making rigid control mechanisms insufficient (Turner, 2014). As a result, leadership research in project management has progressively shifted toward more relational and people-centered perspectives.

Recent studies emphasize that effective project leadership extends beyond technical competence and procedural expertise to include interpersonal skills, emotional awareness, and the ability to motivate and engage team members (Huemann & Turner, 2024) . In dynamic project environments, leaders are required to facilitate collaboration, manage conflicts, and create conditions that enable team members to contribute effectively. This shift reflects a broader transformation in management theory, which increasingly recognizes organizations as social systems rather than purely technical ones.

Furthermore, the rise of temporary and project-based organizations has reinforced the importance of leadership as a social influence process. Unlike traditional hierarchical structures, project teams

often lack stable authority relationships, requiring leaders to rely more on influence, trust, and communication than on formal power (Aaltonen & Kujala, 2016) Consequently, leadership in project management has evolved toward a more human-centered orientation, focusing on relationships, shared goals, and team well-being.

1.1.2 Human-Centered and Inclusive Leadership Approaches

Human-centered leadership refers to leadership approaches that prioritize the needs, experiences, and development of individuals within teams. Rather than viewing team members solely as resources for task execution, human-centered leaders recognize them as active contributors whose motivation, emotions, and well-being significantly affect performance (Van Knippenberg & Sitkin, 2013) . In project management contexts, this perspective is particularly relevant due to the temporary nature of teams and the intensity of collaboration required. Several leadership theories align with human-centered principles. Transformational leadership, for instance, emphasizes inspiring and motivating followers by articulating a shared vision, providing individualized consideration, and fostering intellectual stimulation (Bass & Riggio, 2006) Empirical studies have shown that transformational leadership is positively associated with team motivation, trust, and performance in project environments (Akhavan Tabassi et al., 2025).

Servant leadership represents another influential human-centered approach. Servant leaders focus on supporting and empowering team members, removing obstacles, and facilitating personal growth (Bradley, 1999) In project teams, servant leadership has been linked to higher levels of trust, psychological safety, and collaboration, as leaders prioritize team needs over hierarchical control (Eva et al., 2019)

Ethical and inclusive leadership approaches further contribute to the human-centered paradigm. Ethical leadership emphasizes fairness, transparency, and integrity, which are critical for building trust in project teams operating under time pressure and uncertainty (Brown et al., 2005) Inclusive leadership, meanwhile, focuses on valuing diversity, encouraging participation, and ensuring that team members feel respected and heard (Randel et al., 2018) . Such practices are particularly important in hybrid and diverse project teams, where differences in background, location, and communication styles can create barriers to collaboration.

Emotional intelligence is a central component across these leadership approaches. Leaders with high emotional intelligence are better equipped to recognize and manage their own emotions as well as those of others, enabling more effective communication and conflict management (Goleman et al., 2013). Research indicates that emotionally intelligent project leaders are more successful in fostering motivation, engagement, and psychological safety within their teams (Müller & Turner, 2010).

Overall, human-centered and inclusive leadership approaches shift the focus of project leadership from task control to relationship management, emphasizing the leader's role in creating a supportive environment that enables collaboration and sustained performance.

1.1.3 Leadership Challenges in Hybrid Project Teams

While human-centered leadership offers significant benefits, its implementation becomes more challenging in hybrid project teams. Hybrid teams, characterized by a combination of remote and co-located work, alter traditional patterns of interaction and reduce opportunities for informal communication (Waizenegger et al., 2020). As a result, leaders must adapt their practices to maintain visibility, trust, and engagement across physical and virtual spaces.

One major challenge concerns trust-building. Trust in project teams is often developed through repeated interactions, informal exchanges, and shared experiences. In hybrid environments, reduced face-to-face contact can slow trust development and increase the risk of misunderstandings (Carter et al., 2020). Leaders must therefore adopt deliberate strategies, such as regular check-ins and transparent communication, to compensate for the lack of physical proximity.

Another challenge relates to empowerment and autonomy. Hybrid work arrangements often grant team members greater flexibility and independence, which can enhance motivation but also create ambiguity regarding roles and expectations. Human-centered leaders must balance autonomy with clarity by setting clear goals while allowing team member's discretion in how tasks are performed (Contreras et al., 2020).

Additionally, leaders in hybrid project teams face difficulties in monitoring team dynamics and addressing conflicts. Conflicts may remain hidden longer in virtual settings, and non-verbal cues that signal disengagement or tension are harder to detect. Consequently, project leaders must rely more heavily on communication skills and emotional awareness to identify and address emerging issues before they escalate (AM, 2025)

These challenges highlight the need for a nuanced understanding of how human-centered leadership can be effectively enacted in hybrid project contexts. While existing literature acknowledges the importance of leadership in such environments, empirical insights into how project managers adapt their leadership practices to hybrid settings remain limited. Addressing this gap is essential for understanding how leadership can support team dynamics and collaboration in contemporary project management.

1.2 Communication in Hybrid Project Teams

1.2.1 Role of Communication in Project Success

Communication has long been recognized as a central determinant of project success. In project environments characterized by task interdependence, time pressure, and multiple stakeholders, communication functions as a key coordination and alignment mechanism that enables information sharing, decision-making, and collective action (Institute, 2000) Effective communication allows project teams to establish shared understanding regarding objectives, roles, timelines, and expectations, thereby reducing ambiguity and the risk of errors.

In traditional co-located project teams, communication often occurs through a combination of formal mechanisms such as meetings, reports, and documentation and informal interactions, including spontaneous conversations and social exchanges. These informal interactions play an important role in reinforcing relationships, building trust, and enabling rapid problem-solving (Methot et al., 2026) Research suggests that such informal communication channels contribute significantly to coordination efficiency and team cohesion in project settings.

From a project management perspective, communication is also a critical leadership tool. Project managers rely on communication to articulate vision, align stakeholders, manage expectations, and motivate team members (Huemann & Turner, 2024). Leadership communication influences not only the transmission of information but also how messages are interpreted and acted upon by

team members. Clear, consistent, and transparent communication has been associated with higher levels of trust, commitment, and engagement within project teams (Men & Yue, 2019).

Stakeholder communication represents another essential dimension of project communication. Projects typically involve internal and external stakeholders with diverse interests, levels of power, and expectations. Effective communication helps project managers balance these interests, manage conflicts, and maintain stakeholder support throughout the project lifecycle (Aaltonen & Kujala, 2016). Inadequate communication, by contrast, has been identified as a leading cause of project failure, often resulting in misunderstandings, rework, and delays (Institute, 2000) .

Overall, communication serves both instrumental and relational functions in project teams. Instrumentally, it enables task coordination and information exchange. Relationally, it supports trust-building, social integration, and collaboration. These dual functions become particularly salient in hybrid project teams, where communication is increasingly mediated by digital technologies.

1.2.2 Communication Challenges in Hybrid and Remote Teams

Hybrid project teams fundamentally alter communication patterns by reducing physical co-presence and increasing reliance on digital communication tools. While technologies such as email, instant messaging platforms, and video conferencing facilitate connectivity across distance, they also introduce new challenges that can undermine communication effectiveness (Waizenegger et al., 2020) .

One prominent challenge in hybrid teams is digital communication overload. Team members may receive large volumes of messages across multiple platforms, leading to information fragmentation, reduced attention, and delayed responses (Mazmanian et al., 2013) In project environments where timely information exchange is critical, such overload can impair decision-making and coordination. Moreover, the absence of clear communication norms may exacerbate confusion regarding which channels to use for specific types of information.

Misinterpretation and information loss represent additional challenges. Digital communication often lacks non-verbal cues such as tone, facial expressions, and body language, which play an important role in conveying meaning and emotional intent (Daft & Lengel, 1986) As a result, messages may be interpreted differently than intended, increasing the likelihood of

misunderstandings and conflict. This risk is heightened in hybrid teams composed of members from diverse cultural and professional backgrounds.

Time zone differences and asynchronous work further complicate communication in hybrid project teams. When team members work in different locations and time zones, opportunities for real-time interaction are limited. Although asynchronous communication offers flexibility, it can slow feedback cycles and hinder rapid problem resolution (Dannals et al., 2021) Delays in communication may accumulate over time, negatively affecting project schedules and team morale.

Additionally, hybrid teams often experience reduced informal communication. The lack of spontaneous interactions such as hallway conversations or informal check-ins can weaken social bonds and make it more difficult for team members to raise concerns or share tacit knowledge (Carter et al., 2020) This reduction in informal exchanges may contribute to feelings of isolation and disengagement, particularly among remote team members.

These challenges suggest that communication in hybrid project teams requires deliberate management. Simply transferring traditional communication practices to digital platforms is often insufficient. Instead, project managers must adopt communication strategies that address the structural and social complexities introduced by hybrid work arrangements.

1.3 Human-Centred Communication Practices

In response to the challenges associated with hybrid work, recent literature emphasizes the importance of human-centred communication practices. Human-centred communication focuses not only on the efficient transmission of information but also on the relational and emotional aspects of interaction, recognizing communication as a social process that shapes trust, inclusion, and collaboration (Men & Yue, 2019) .

Active listening and feedback are central elements of human-centred communication. Active listening involves attentively engaging with others' perspectives, asking clarifying questions, and demonstrating empathy. In project teams, leaders who practice active listening are better positioned to understand team members' concerns, identify emerging issues, and foster a sense of psychological safety (A. C. Edmondson & Lei, 2014) Constructive feedback, when delivered respectfully and transparently, further supports learning and continuous improvement.

Transparency and shared understanding are also critical in hybrid project teams. Transparent communication involves openly sharing information about project goals, decisions, and changes, thereby reducing uncertainty and speculation. Research indicates that transparency enhances trust and commitment, particularly in environments where team members have limited visibility into others' work activities (Carter et al., 2020). Shared understanding, often referred to as "shared mental models," enables team members to coordinate effectively by aligning their interpretations of tasks and roles (Mathieu et al., 2000) .

Inclusive communication practices play a particularly important role in hybrid contexts. Inclusive communication ensures that all team members, regardless of location or status, have opportunities to contribute and be heard. This may involve deliberately inviting input from remote participants during meetings, rotating facilitation roles, or using asynchronous tools to gather perspectives (Randel et al., 2018). Inclusive communication practices help mitigate power imbalances and reduce the marginalization of remote team members.

Finally, leadership communication is central to sustaining collaboration in hybrid project teams. Leaders set the tone for communication by modelling openness, respect, and empathy. Studies suggest that leaders who communicate consistently, acknowledge uncertainty, and demonstrate concern for team well-being are more effective in maintaining engagement and cohesion in hybrid teams (Contreras et al., 2020) .

In summary, human-centred communication practices provide a critical foundation for effective collaboration in hybrid project teams. By addressing both task-related and relational dimensions of communication, such practices help project managers navigate the challenges of hybrid work and support positive team dynamics.

1.4 Team Dynamics and Collaboration in Project Teams

1.4.1 Understanding Team Dynamics in Project Environments

Team dynamics refer to the patterns of interaction, relationships, and behaviours that emerge among members of a team as they work together toward shared objectives. In project environments, team dynamics are particularly important due to the temporary nature of teams, high task interdependence, and pressure associated with deadlines and performance expectations (Mathieu et al., 2017). Unlike permanent organizational teams, project teams are often formed for

a limited duration, bringing together individuals with diverse skills, backgrounds, and professional identities.

Classical theories of group development provide a useful foundation for understanding team dynamics. Tuckman's model of group development, which outlines the stages of forming, storming, norming, performing, and adjourning, has been widely applied in project management research (Tuckman & Jensen, 1977). Although the linear progression suggested by this model has been critiqued, it remains valuable in illustrating how teams evolve over time and how conflict, cohesion, and performance are interrelated.

In project teams, roles, norms, and interdependencies strongly influence team dynamics. Clear role definition helps team members understand their responsibilities and how their tasks contribute to overall project goals, thereby reducing ambiguity and coordination breakdowns (Belbin, 2010). Norms related to communication, decision-making, and conflict resolution further shape how team members interact and collaborate. When roles and norms are poorly defined, project teams are more likely to experience tension, duplication of effort, and reduced efficiency.

Interdependence is another defining feature of project teams. Tasks are often sequential or reciprocal, meaning that the output of one team member becomes the input for another. High interdependence increases the need for coordination and mutual adjustment, making effective team dynamics essential for project success (Thompson, 1967). In hybrid project teams, where interactions are mediated by digital tools, managing these interdependencies becomes even more complex, highlighting the importance of leadership and communication practices that support collaboration.

1.4.2 Trust, Motivation, and Psychological Safety

Trust is widely regarded as a foundational element of effective team dynamics and collaboration. In project teams, trust enables members to rely on one another's competence and intentions, facilitating information sharing and cooperative behaviour (Costa et al., 2018). Trust is particularly critical in hybrid teams, where reduced visibility and limited face-to-face interaction can make it harder to assess others' actions and commitment.

Research distinguishes between cognitive trust, based on perceptions of competence and reliability, and affective trust, grounded in interpersonal relationships and emotional bonds

(McAllister, 1995) Both forms of trust contribute to collaboration in project teams. Cognitive trust supports task coordination, while affective trust enhances willingness to share concerns, admit mistakes, and provide mutual support. Studies indicate that leadership behaviours and communication practices play a central role in fostering both types of trust, especially in virtual and hybrid settings (Carter et al., 2020)

Motivation is another key component of team dynamics in project environments. Project-based work often involves high workloads, uncertainty, and tight deadlines, which can strain individual motivation over time. Motivation in project teams is influenced not only by extrinsic factors such as rewards and recognition but also by intrinsic factors, including autonomy, meaningful work, and a sense of belonging (Ryan & Deci, 2000) Human-centred leadership and communication practices can enhance intrinsic motivation by acknowledging individual contributions and supporting autonomy.

Psychological safety has emerged as a critical concept linking trust, motivation, and learning in teams. Psychological safety refers to a shared belief that the team is safe for interpersonal risk-taking, such as speaking up, asking questions, or challenging existing practices (A. Edmondson, 1999) In project teams, psychological safety enables members to raise concerns early, share innovative ideas, and address problems before they escalate. Empirical research demonstrates that psychological safety is positively associated with team learning, adaptability, and performance, particularly in complex and uncertain environments (A. C. Edmondson & Lei, 2014) .

In hybrid project teams, maintaining psychological safety can be challenging due to reduced informal interaction and reliance on mediated communication. Leaders and project managers play a crucial role in signalling openness and respect, encouraging participation, and responding constructively to input from team members. Without psychological safety, hybrid teams may experience silence, disengagement, and unresolved conflicts, undermining collaboration and project outcomes.

1.4.3 Conflict and Collaboration in Hybrid Teams

Conflict is an inevitable aspect of teamwork, particularly in project environments characterized by diversity, interdependence, and time pressure. Rather than viewing conflict solely as a negative phenomenon, contemporary research distinguishes between different types of conflict and their varying effects on team performance (De Dreu & Weingart, 2003) .

Task conflict arises from differences in viewpoints, ideas, or opinions related to the content of the work. Moderate levels of task conflict can be beneficial, as they encourage critical thinking and improve decision quality. Process conflict, by contrast, involves disagreements about how work should be organized, including roles, responsibilities, and resource allocation. Relational conflict refers to interpersonal tensions and emotional disagreements, which are generally detrimental to collaboration and team well-being.

In hybrid project teams, conflict dynamics are shaped by communication constraints and reduced social cues. Misunderstandings may persist longer due to asynchronous communication, and relational conflicts may be harder to detect and resolve without face-to-face interaction (Dannals et al., 2021). Cultural diversity and differing communication styles can further intensify conflict if not managed effectively.

Conflict management approaches play a critical role in determining whether conflict undermines or enhances collaboration. Constructive conflict management involves open communication, mutual respect, and a focus on shared goals rather than personal interests (DeChurch et al., 2013). Leaders who encourage dialogue and facilitate problem-solving can help teams leverage task conflict while minimizing relational tension.

Collaboration in project teams emerges from the effective management of team dynamics, including trust, motivation, and conflict. Collaboration involves coordinated action, shared responsibility, and collective problem-solving, enabling teams to integrate diverse expertise and respond to challenges effectively (Kozlowski & Bell, 2013). In hybrid environments, collaboration depends heavily on leadership and communication practices that support alignment, inclusion, and continuous interaction.

Overall, team dynamics and collaboration are deeply intertwined in project teams. Understanding how trust, motivation, psychological safety, and conflict interact provides a foundation for examining how human-centred leadership and communication practices influence collaboration in hybrid project settings. This perspective reinforces the need for empirical investigation into the lived experiences of project managers and team members navigating these complex dynamics.

1.5 Hybrid Project Teams as a Contemporary Work Context

1.5.1 Emergence of Hybrid Work in Project Management

Hybrid work has emerged as a dominant work arrangement in many organizations, particularly in project-based environments. Advances in digital technologies, globalization, and the growing emphasis on flexibility and work–life balance had already begun to reshape work practices prior to 2020. However, the COVID-19 pandemic significantly accelerated the adoption of hybrid and remote work models, transforming them from temporary solutions into long-term organizational strategies (Kniffin et al., 2021) .

In project management contexts, hybrid work refers to arrangements in which project team members alternate between working remotely and working from shared physical locations. This model enables organizations to access geographically dispersed talent, reduce operational costs, and offer greater flexibility to employees. At the same time, it alters traditional assumptions about coordination, supervision, and collaboration within project teams (Dannals et al., 2021) .

Project-based work is particularly well suited to hybrid arrangements due to its outcome-oriented nature. Projects are typically defined by clear objectives, timelines, and deliverables, which allows work to be structured around outputs rather than physical presence. As a result, many organizations have integrated hybrid work into project governance structures, relying on digital tools to manage communication, documentation, and performance (Institute, 2000)

Despite these advantages, the transition to hybrid work represents a profound shift in how project team’s function. Research indicates that hybrid teams experience both benefits and limitations. On the positive side, hybrid work can enhance autonomy, job satisfaction, and perceived work–life balance (Wang et al., 2021) On the negative side, it can weaken social cohesion, reduce informal learning, and increase the risk of communication breakdowns if not managed effectively.

These tensions highlight the need to reconsider traditional project management practices. While methodologies and tools remain important, the success of hybrid project teams increasingly depends on how leaders and team members navigate social interaction, communication norms, and shared responsibility across physical and virtual spaces.

1.6 Managerial Implications of Hybrid Project Environments

The rise of hybrid project teams has significant managerial implications, particularly for project managers responsible for coordination, control, and team performance. One of the most prominent challenges concerns coordination and control. In hybrid environments, project managers have less direct visibility into daily work activities, making it more difficult to monitor progress and detect issues early (Wang et al., 2021). This shift requires a move away from presence-based control toward trust-based and outcome-oriented management.

Leadership adaptation is therefore essential in hybrid project settings. Project managers must develop new capabilities related to virtual leadership, including the ability to build trust at a distance, communicate effectively through digital channels, and maintain team engagement without relying on physical proximity (Contreras et al., 2020). Research suggests that leaders who fail to adapt their style may experience reduced team commitment and increased disengagement among remote team members.

Communication structures also require careful redesign in hybrid project teams. Informal communication, which traditionally played a key role in coordination and relationship-building, is less likely to occur spontaneously in hybrid settings. Project managers must therefore create intentional communication routines, such as regular check-ins, structured meetings, and clear documentation practices, to ensure information flows effectively (Waizenegger et al., 2020). Without such structures, hybrid teams risk fragmentation and misalignment.

Another important implication relates to equity and inclusion. Hybrid work arrangements can unintentionally create disparities between team members who are physically present more often and those who work remotely. Research indicates that remote team members may feel less visible and less involved in decision-making processes, which can undermine motivation and collaboration (Raghuram et al., 2019). Inclusive leadership and communication practices are therefore critical to ensuring that all team members have equal opportunities to contribute and be recognized.

Finally, hybrid project environments place greater emphasis on human-centred management practices. As discussed in previous sections, trust, psychological safety, and motivation are central to effective collaboration. In hybrid contexts, these elements cannot be taken for granted and must be actively cultivated through leadership behaviour and communication. Project managers who

adopt human-centred approaches are better positioned to address the social and relational challenges of hybrid work, thereby enhancing team dynamics and collaboration.

In summary, hybrid project teams represent a contemporary work context that reshapes traditional project management assumptions. While hybrid work offers flexibility and efficiency, it also introduces new challenges related to coordination, communication, and leadership. Understanding these implications provides essential context for examining how human-centred leadership and communication practices can support collaboration in hybrid project teams, reinforcing the relevance of the empirical study that follows.

1.7 Synthesis of the Literature and Identification of Research Gaps

The review of the scientific literature highlights a clear evolution in project management research toward a more human-centred perspective. Across the examined streams of literature, leadership, communication, and team dynamics consistently emerge as critical factors influencing collaboration and project success. Contemporary leadership research emphasizes the shift from task-oriented control toward relational, inclusive, and people-centred leadership approaches that prioritize trust, motivation, and psychological safety. Such approaches are increasingly recognized as essential in complex and dynamic project environments, particularly where formal authority is limited and collaboration is critical.

Similarly, communication literature underscores its dual role in project teams: as a functional mechanism for coordination and information exchange, and as a relational process that shapes trust, engagement, and shared understanding. In hybrid project teams, communication challenges related to digital overload, misinterpretation, and reduced informal interaction are widely documented. Research suggests that human-centred communication practices such as active listening, transparency, and inclusive participation can mitigate these challenges and support collaboration, yet their practical implementation remains underexplored.

The literature on team dynamics further reinforces the importance of trust, motivation, psychological safety, and constructive conflict management in enabling effective collaboration. While studies acknowledge that these elements are interdependent and influenced by leadership and communication, much of the existing research examines them in isolation or within traditional co-located teams. Hybrid project teams, by contrast, introduce additional complexity due to reduced physical proximity, reliance on digital tools, and altered social interaction patterns.

Despite the growing body of research on hybrid work, several gaps remain evident. First, there is limited empirical research that integrates human-centred leadership, communication practices, and team dynamics within a single analytical framework focused on hybrid project teams. Second, while theoretical models highlight the importance of trust, psychological safety, and collaboration, fewer studies examine how project managers enact these principles in practice within hybrid environments. Third, much of the existing literature remains conceptual or prescriptive, with insufficient qualitative insight into the lived experiences of project managers navigating hybrid team challenges.

These gaps point to the need for exploratory empirical research that captures how human-centred leadership and communication practices are experienced, interpreted, and applied in hybrid project contexts. Addressing these gaps can contribute to a more nuanced understanding of collaboration in contemporary project management and provide actionable insights for practitioners.

1.8 Summary

This chapter has reviewed and synthesized key streams of scientific literature related to human-centred leadership, communication in project teams, team dynamics and collaboration, and hybrid project environments. The analysis demonstrates that while leadership and communication are widely recognized as central to project success, their role becomes even more critical in hybrid work settings, where traditional coordination mechanisms and social interactions are disrupted.

The literature reveals a clear consensus on the importance of trust, motivation, psychological safety, and inclusive practices for effective collaboration. However, it also highlights significant gaps in empirical understanding, particularly regarding how project managers apply human-centred leadership and communication practices in hybrid project teams. These gaps justify the development of an analytical framework that integrates leadership, communication, team dynamics, and collaboration.

In response to these findings, the next chapter develops the analytical framework that guides this research. Chapter 2 defines the key constructs and relationships derived from the literature and formulates research propositions that structure the empirical investigation presented in Part 2 of the thesis.

CHAPTER 2

2 DEVELOPMENT OF THE ANALYTICAL FRAMEWORK

2.1 Introduction to Chapter 2

Chapter 1 demonstrated that contemporary project management research increasingly emphasizes human-centred leadership, communication practices, and team dynamics as critical drivers of collaboration, particularly in hybrid project environments. The literature review highlighted that while these concepts are widely studied, they are often examined in isolation or within traditional co-located teams. As a result, there is limited integrated understanding of how leadership and communication jointly shape team dynamics and collaboration in hybrid project teams.

In response to this gap, the purpose of this chapter is to structure the theoretical insights identified in Chapter 1 into a coherent analytical framework. This framework serves as a conceptual foundation for the empirical study by clarifying the key constructs of the research and establishing the relationships between them. Specifically, this chapter aims

(1) to define and operationalize the core concepts of human-centred leadership, communication, team dynamics, and collaboration, and (2) to explain how these concepts interact within hybrid project teams.

2.2 Definition and Operationalization of Key Concepts

2.2.1 Human-Centred Leadership

Human-centred leadership refers to leadership approaches that prioritize individuals' needs, well-being, and development alongside task achievement. Recent leadership research emphasizes that effective leaders focus not only on performance outcomes but also on empathy, ethical behaviour, and the creation of supportive work environments (Bugarčić & Slavković, 2023). In project management contexts, where teams are temporary and interdependent, such leadership is particularly relevant.

The literature identifies several core dimensions of human-centred leadership. **Empathy and emotional intelligence** enable leaders to recognize and respond to team members' emotional states, which is essential for maintaining engagement and psychological safety (Miao et al., 2017). **Empowerment and autonomy** involve granting team members discretion in how they perform

tasks, thereby fostering intrinsic motivation and ownership (Breevaart & Zacher, 2019) .**Trust-building and ethical behaviour** contribute to perceptions of fairness and reliability, which are crucial in project teams operating under uncertainty(Hoch et al., 2018).

In hybrid project teams, the role of project managers as human-centred leaders becomes even more critical. Reduced face-to-face interaction increases reliance on leadership behaviours communicated digitally, making empathy, clarity, and ethical conduct central to sustaining trust and collaboration(Contreras et al., 2020).

2.3 Communication in Hybrid Project Teams

Communication in project contexts refers to the processes through which information, meaning, and understanding are exchanged among team members and stakeholders. In hybrid project teams, communication is predominantly mediated by digital tools and occurs through both synchronous and asynchronous channels (Waizenegger et al., 2020) .

The literature identifies several key dimensions of effective communication. **Communication clarity** ensures that messages are understood as intended, reducing ambiguity and rework(Institute, 2000). **Frequency and feedback** support alignment and enable timely correction of issues, particularly in fast-paced project environments (Dannals et al., 2021). **Transparency and inclusiveness** involve openly sharing information and ensuring that all team members, regardless of location, can participate in discussions and decision-making processes (Randel et al., 2018).

Human-centred communication practices emphasize not only efficiency but also relational quality. Active listening, respectful dialogue, and responsiveness contribute to trust and psychological safety, which are essential for collaboration in hybrid teams (Men & Yue, 2019). Without such practices, communication breakdowns may persist and undermine team dynamics.

2.4 Team Dynamics in Hybrid Project Teams

Team dynamics refer to the patterns of interaction, relationships, and psychological states that develop within a team over time. In project environments, team dynamics are shaped by task interdependence, time pressure, and diversity of expertise (Mathieu et al., 2017).

Key elements of team dynamics include **trust and psychological safety, motivation and engagement, and conflict and conflict management**. Trust enables cooperation and knowledge sharing, while psychological safety encourages speaking up and learning behaviours (A. C.

Edmondson & Lei, 2014) Motivation and engagement influence the level of effort and commitment team members bring to the project. Conflict, when managed constructively, can enhance decision quality, but unmanaged relational conflict can damage collaboration (DeChurch et al., 2013).

Hybrid work introduces specific challenges to team dynamics by reducing informal interaction and increasing dependence on mediated communication. As a result, team dynamics in hybrid project teams are particularly sensitive to leadership behaviour and communication quality (Carter et al., 2020).

2.5 Collaboration in Project Teams

Collaboration refers to the extent to which team members coordinate their efforts, share knowledge, and jointly solve problems to achieve shared goals. In project management, collaboration is a critical outcome that enables teams to integrate diverse expertise and respond effectively to uncertainty (Kozlowski & Bell, 2013).

Indicators of effective collaboration include coordination and alignment, shared understanding, and collective problem-solving. Collaborative teams demonstrate mutual accountability and adaptability, allowing them to manage interdependencies and deliver project outcomes efficiently (Guide, 2008). In this study, collaboration is conceptualized as an outcome influenced by leadership, communication, and team dynamics rather than as an isolated behaviour.

Table 1 Summary of Key Constructs and Dimensions

Construct	Core Dimensions
Human-Centred Leadership	Empathy, emotional intelligence, empowerment, trust, ethical behaviour
Communication	Clarity, feedback, transparency, inclusiveness
Team Dynamics	Trust, psychological safety, motivation, conflict management
Collaboration	Coordination, shared understanding, collective problem-solving

2.6 Relationships between Key Concepts

2.6.1 Influence of Human-Centred Leadership on Team Dynamics

Human-centred leadership has been shown to significantly influence team dynamics. Leadership behaviours that emphasize empathy, fairness, and support contribute to trust and psychological safety, enabling team members to engage openly and collaboratively. Empowering leadership practices enhance motivation by fostering autonomy and ownership, while ethical leadership reduces relational conflict and strengthens cohesion.(Hoch et al., 2018)

In hybrid project teams, leadership influence is amplified due to reduced informal interaction. Project managers' leadership behaviours become key signals shaping perceptions of trust, engagement, and conflict management (Contreras et al., 2020).

2.6.2 Role of Communication in Shaping Team Dynamics

Communication practices directly shape team dynamics by influencing shared understanding, trust, and engagement. Clear and transparent communication reduces ambiguity, while regular feedback supports alignment and learning (Dannals et al., 2021) In hybrid environments, communication also acts as a mediator of trust, compensating for the lack of physical proximity.

However, communication challenges such as digital overload and misinterpretation can undermine team dynamics if not addressed through human-cantered practices (Waizenegger et al., 2020) .

2.6.3 Joint Influence of Leadership and Communication on Collaboration

Leadership and communication interact synergistically to influence collaboration. Leadership intent is largely conveyed through communication, making alignment between leadership style and communication practices essential. When human-centred leadership is supported by inclusive and transparent communication, team dynamics are strengthened, leading to higher levels of collaboration (Men & Yue, 2019) .

2.7 Research Propositions

In line with the qualitative and exploratory nature of this study, the following research propositions are formulated:

Proposition 1: Human-centred leadership positively influences team dynamics in hybrid project teams.

Proposition 2: Human-centred communication practices positively influence team dynamics in hybrid project teams.

Proposition 3: Positive team dynamics enhance collaboration within hybrid project teams.

These propositions guide data collection and analysis without implying statistical testing.

2.8 Conclusion

This chapter developed the analytical framework that underpins the empirical study. By defining key constructs and explaining their relationships, it provides a structured lens for examining how human-centred leadership and communication practices influence team dynamics and collaboration in hybrid project teams. The framework establishes a clear link between theory and empirical inquiry, guiding the qualitative methodology presented in the next part of the thesis. Part 2 begins with Chapter 3, which outlines the research methodology adopted to investigate these relationships.

PART 2

EMPIRICAL STUDY

2.9 Introduction to Part 2

The objective of Part 2 of this thesis is to empirically investigate how human-centered leadership and communication practices influence team dynamics and collaboration in hybrid project teams. While Part 1 established the theoretical foundations of the study and developed an analytical framework linking leadership, communication, team dynamics, and collaboration, Part 2 seeks to explore how these relationships manifest in practice within contemporary project environments.

The analytical framework developed in Chapter 2 serves as the guiding lens for the empirical investigation. Specifically, it informs the selection of the research design, the development of the data collection instruments, and the structure of the data analysis. Human-centered leadership and communication are examined as key managerial practices, team dynamics are considered as an intermediate mechanism, and collaboration is analyzed as a central outcome within hybrid project teams. This framework ensures coherence between the theoretical insights derived from the literature and the empirical data collected from practitioners.

Given the exploratory nature of the research problem and the focus on understanding experiences, perceptions, and practices, a qualitative research approach was selected. Qualitative methods are particularly suitable for studying complex social and organizational phenomena where context, interaction, and meaning play a central role (Creswell & Poth, 2016). In the context of hybrid project teams, qualitative inquiry enables an in-depth exploration of how project managers and team members perceive leadership behaviors, communication processes, and their effects on team dynamics and collaboration.

By adopting a qualitative approach, this study aims to capture rich, contextualized insights that cannot be fully accessed through quantitative measures alone. This empirical investigation therefore complements the theoretical analysis presented in Part 1 and contributes to a deeper understanding of human-centered management practices in hybrid project settings. The following chapter presents the research methodology employed to conduct this empirical study.

CHAPTER 3

3 RESEARCH METHODOLOGY

3.1 Introduction

Research methodology refers to the structured framework that directs the research process. It encompasses the instruments, methodologies, and strategies used by researchers to gather, analyze, and evaluate data to investigate a particular research issue. It serves as a blueprint for executing the study in a methodologically rigorous and systematic way, confirming that the research topic is properly resolved.

In this research, a qualitative research methodology was adopted to examine human-centered leadership, communication processes, and cross-functional collaboration in automotive automation projects that function in hybrid work environments. Qualitative methodologies are particularly appropriate for examining complex social and organizational phenomena where meaning, context, and interpretation are central (Creswell & Puth, 2021). The choice of a

qualitative research methodology was informed by the need to delve deeper into the experiences, perceptions, and interpretations of the participants regarding their experiences and understanding of leadership processes, team processes, and coordination mechanisms in complex and technology-intensive project environments. Through the use of in-depth semi-structured interviews, the research aimed to delve deeper into the experiences of the participants and provide rich data that is context-specific and highlights the role of human-centered factors such as trust, motivation, role perception, and adaptive managerial processes in cross-functional collaboration and project outcomes in cross-functional and distributed teams.

The chapter is structured as follows. First, the research design and methodological approach are outlined. Next, the study context and sampling strategy are presented. The data collection methods and data analysis procedure are then described. Finally, ethical considerations guiding the research are discussed.

3.2 Research Procedure

This research adopted a qualitative methodology, in which semi-structured interviews were employed as the primary data collection method. This approach was considered most suitable as it enabled the researcher to gain insights into the experiences and perceptions of professionals involved in cross-functional automotive automation projects operating in hybrid work environments (Hennink et al., 2020). In this study, the questions aimed to explore participants' perspectives on human-centered leadership, communication practices, cross-functional coordination, team dynamics, and managerial strategies that influence collaboration and adaptability in hybrid automotive projects.

In figure 1, a brief of the research procedure is depicted.

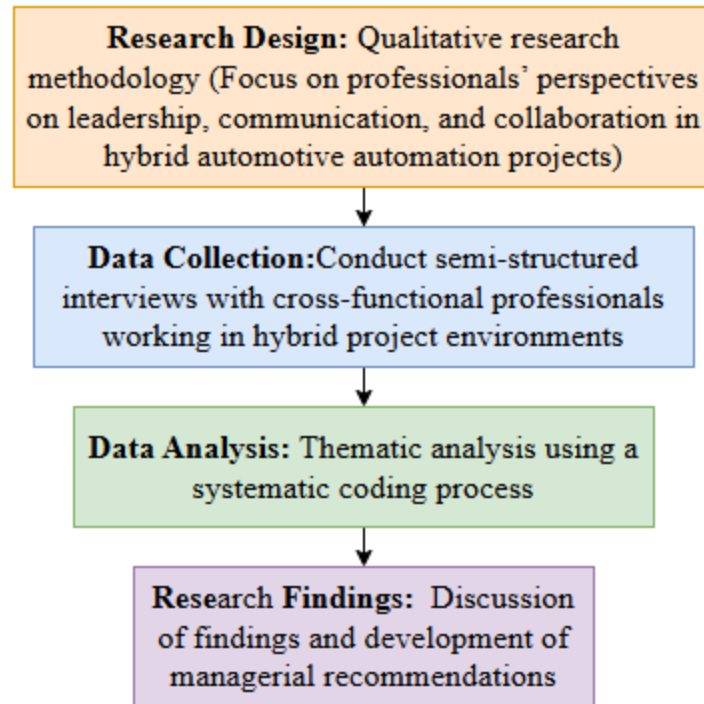


Figure 1 Research Procedure Flowcharts

3.3 Research Design

Given the exploratory nature of this study, a qualitative approach using semi-structured interviews was deemed most suitable. Qualitative research is particularly appropriate for investigating human-centered factors, as it allows for a deep understanding of participants' experiences, perceptions, and interpretations within complex, technology-intensive environments. Semi-structured interviews are widely recognized as an effective method for collecting rich, context-specific data while providing the flexibility to explore emerging insights.

Face-to-face and virtual interviews were employed, allowing for consistency across participants while accommodating the hybrid work context of the study. This approach enabled participants to provide detailed accounts of their experiences with leadership, communication, and cross-functional collaboration, and allowed the interviewer to probe deeper into specific responses when necessary.

The development of the interview questions was guided by key concepts related to human-centered leadership, communication processes, team dynamics, and managerial practices in hybrid

automotive automation projects. The semi-structured format enabled the exploration of participants' perspectives on leadership behaviors, coordination mechanisms, and collaborative practices, as well as challenges and opportunities arising from hybrid work arrangements.

The semi-structured interview questions were organized around five thematic areas:

- Human-Centered Leadership in Cross-Functional Automotive Projects – to explore leadership styles, trust-building, motivation, and empowerment practices.
- Communication Practices and Information Flow in Hybrid Teams – to examine formal and informal communication mechanisms, knowledge sharing, and challenges in distributed work.
- Cross-Functional Coordination and Role Alignment – to investigate role clarity, decision-making responsibilities, and alignment across functions.
- Team Dynamics and Conflict Management in Hybrid Environments – to understand conflict types, resolution strategies, and interpersonal dynamics.
- Managerial Practices for Enhancing Collaboration and Adaptability – to capture practices that promote adaptability, resilience, and overall team effectiveness.

The semi-structured interview guide allowed for probing follow-up questions, ensuring that the study could capture nuanced perspectives on leadership and collaboration practices in hybrid automotive projects. The design ensured that data collected were directly aligned with the research objectives while allowing sufficient flexibility to uncover insights not initially anticipated by the researcher.

Sample interview questions included:

- How would you describe the leadership approach used in your current or most recent project?
- How communication is typically structured and managed within your cross-functional project team?
- How are roles and responsibilities defined and coordinated across different functions in your project?
- What types of conflicts tend to arise in hybrid project teams, and how are they managed?

- What managerial practices or behaviors have proven most effective in enhancing collaboration and adaptability?

This research design ensures that the study systematically explores the human-centered elements influencing cross-functional collaboration, communication, and leadership in hybrid automotive automation projects, providing a comprehensive understanding of best practices and challenges in complex technological environments.

paraphrase

3.4 Sampling Technique

For the qualitative component, purposive sampling was employed to select participants with extensive professional experience in automotive automation projects, cross-functional collaboration, and hybrid work environments. This method was deemed appropriate because it allowed the selection of information-rich cases and ensured that the participants had the necessary expertise to respond to the research objectives. The qualitative data gathering involved conducting 15 semi-structured in-depth interviews with experienced professionals in key functional areas such as product management, engineering, quality assurance, and operations in automotive companies involved in automation and digital transformation initiatives. The participants were selected from both domestic and multinational companies operating in technologically intensive project environments. The sample size was determined based on the principle of data saturation, whereby no new themes or insights emerged during the later stages of data collection. This ensured methodological rigor while maintaining efficiency. The application of purposive sampling made it possible to gather detailed experience-driven data that was directly related to the theme of human-centered leadership, communication, and cross-functional coordination in hybrid automotive automation projects.

3.5 Data Collection

The data gathering technique is crucial to the success of research since it makes the obtained data correctly accessible, which makes it easier to evaluate the study's findings. This study includes both primary and secondary data collecting, as well as primary surveys, interviews, and secondary thematic analysis. Thus, the technique of gathering data is essential to carrying out a fruitful research study (Ummah, 2019).

Data for this study were collected primarily through in-depth semi-structured interviews with

professionals involved in cross-functional automotive automation projects operating in hybrid work environments. This method was selected to capture rich, experience-based insights into human-centered leadership, communication practices, coordination mechanisms, and team dynamics. The interviews were conducted either face-to-face or virtually, depending on participant availability and geographical constraints, reflecting the hybrid nature of contemporary project work. An interview guide structured around five thematic areas leadership, communication, cross-functional coordination, team dynamics, and managerial practices was used to ensure consistency while allowing flexibility for probing and follow-up questions. All interviews were conducted with informed consent, recorded with participant permission, and later transcribed verbatim for analysis. In addition to primary data, relevant secondary sources such as academic journals, industry reports, and organizational documents were reviewed to contextualize the findings and support interpretation. This combined approach enabled a comprehensive and methodologically robust exploration of collaboration and leadership practices in hybrid automotive automation projects.

Paraphrase

3.5.1 Interview Guide

An interview guide was developed to ensure consistency across interviews while allowing flexibility for follow-up questions. The guide was structured around key themes derived from the analytical framework, including human-centred leadership, communication practices, team dynamics, and collaboration. The use of an interview guide supports methodological rigor by ensuring alignment between research questions and data collection while preserving the exploratory nature of qualitative inquiry (Tracy, 2020).

The full interview guide is provided in the appendices.

3.6 Data Analysis

Qualitative Analysis

The qualitative aspect of this research is mostly based on data obtained from semi-structured interviews and focus group discussions, analyzed using thematic analysis. This approach entails methodically transcribing dialogues, recognizing repeating patterns, coding segments, and categorizing themes that arise from participant answers. Thematic analysis helps in making sense of intricate findings and behavioral trends by creating meaningful groups of data, thus improving

comprehension of the human-related factors that affect leadership, communication, and cross-functional collaboration in hybrid automotive automation projects. This type of analysis helps in the investigation of subjective experiences, perceptions, and project-related challenges in technologically complex settings.

Interviews

Interviews provide a comprehensive insight into personal perspectives. They possess a semi-structured format, enabling the researcher to explore beyond predetermined questions in response to participant answers. This adaptability facilitates a more comprehensive and intricate data collection. Surveys provide standardized responses that facilitate comparison, but interviews elicit emotions, decision-making processes, and contextual input in participants' own language, rendering them an essential instrument for investigating qualitative aspects (Brooks et al., 2019).

Focus groups

Focus groups enhance the data by facilitating interactive talks among 6 to 10 participants, overseen by a researcher. These meetings enable the examination of group viewpoints, discrepancies, and agreements, often resulting in dynamic discoveries that may not emerge in individual interviews. Plummer (2017) asserts that focus groups may be very effective, particularly when members openly share ideas, contesting and broadening one another's perspectives (Plummer, 2017).

Thematic analysis

Thematic analysis was used as a systematic and reflexive approach to the analysis of qualitative data. The researcher undertook repeated analysis of the interview transcripts to identify meaningful patterns and ideas, which were then coded into themes that represented critical aspects of cross-functional collaboration and leadership. The themes included human-centered leadership, communication practices, cross-functional coordination, team dynamics, and managerial approaches to adaptability. The approach enabled the researcher's interpretive perspective to actively inform the analysis while ensuring that the findings remained grounded in the data. The findings were then synthesized to support conclusions about the role of leadership and communication practices in shaping collaboration, resilience, and performance in hybrid automotive automation projects.

The resulting themes formed the basis for the presentation and discussion of findings in Chapter 4.

3.7 Ethical Considerations

Ethical considerations were central to the research process. All participants were informed about the purpose of the study, the voluntary nature of participation, and their right to withdraw at any time. Informed consent was obtained prior to data collection. Participant anonymity and confidentiality were strictly maintained through the use of pseudonyms and secure data storage.

The study adhered to institutional ethical guidelines and followed established ethical principles for qualitative research involving human participants (British Psychological Society, 2021).

3.8 Summary

This chapter outlined the research methodology adopted to explore human-centered leadership, communication practices, and cross-functional collaboration in hybrid automotive automation projects. A qualitative research design using semi-structured interviews was employed to capture rich, experience-based insights from professionals across key functional areas. Purposive sampling ensured the selection of information-rich participants, while thematic analysis was used to identify recurring patterns and critical themes within the data. The systematic procedures for data collection and analysis provided a robust foundation for interpreting participants' perspectives. The following chapter presents the qualitative results derived from this methodological framework, highlighting key themes and insights.

CHAPTER 4

4 Qualitative Results

This chapter presents the qualitative results of the study, focusing on human-centered factors that shape cross-functional collaboration in automotive automation projects operating within hybrid work environments in automotive automation projects. It provides an in-depth analysis of participants on leadership approaches, communication practices, coordination mechanisms, team dynamics, and managerial strategies that influence collaboration, adaptability and project outcomes. Drawing on insights from experienced professionals across product management, engineering, quality, and operations functions, the chapter explores how now-technical, human-

centered elements contribute to the effectiveness of complex and technologically intensive automotive projects.

The study employs in-depth semi-structured interviews to investigate how leadership behaviors, structures of communication, and interpersonal dynamics shape trust, motivation, alignment, and performance within cross-functional and distributed teams. Particular attention is focused on both challenges and opportunities resulting from hybrid work arrangements, such as issues related to information flow, clarity over roles, conflict management, and adaptive decision-making. The findings highlight how managers and team leaders navigate uncertainty, coordinate across functional boundaries, and foster collaboration in environments characterized by rapid technological change and organizational complexity.

Additionally, this chapter delves deeper into how effective management practices and leadership characteristics ensure that resilience and flexibility are exercised by teams in order to accommodate changing demands and conditions. Moreover, this chapter identifies recommendations given by participants on how product and project managers can implement teamwork between different functions in hybrid projects within the automotive industry.

The findings are organized around five key themes that capture critical dimensions of the collaboration and leadership in hybrid automotive automation projects:

- ❖ **Theme 1:** Human-Centered Leadership in Cross-Functional Automotive Projects
- ❖ **Theme 2:** Communication Practices and Information Flow in Hybrid Teams
- ❖ **Theme 3:** Cross-Functional Coordination and Role Alignment
- ❖ **Theme 4:** Team Dynamics and Conflict Management in Hybrid Environments
- ❖ **Theme 5:** Managerial Practices for Enhancing Collaboration and Adaptability

Each of the theme offers insight into distinct yet complementary perspectives on human-centered project management practices, including leadership style, communication structure, conflict resolution, and adaptive managerial practice. Together, these themes provide a comprehensive understanding of how cross-functional collaboration can be strengthened in hybrid automotive automation projects, contributing to improved team effectiveness, project delivery, and organizational performance.

Table 2 Themes and sub-themes on Human-centered Leadership, Communication, and collaboration in Hybrid Automotive Automation Projects

Themes	Description	Codes
Theme 1: Human-Centered Leadership in Cross-Functional Automotive Projects	Explores leadership approaches and behaviors that influence trust, motivation, and collaboration within cross-functional automotive automation projects, particularly in hybrid work environments.	Leadership Style (collaborative vs. directive), Decision-Making Clarity, Leader Accessibility and Availability, Trust-Building Behaviors, Psychological Safety, Empowerment and Autonomy.
Theme 2: Communication Practices and Information Flow in Hybrid Teams	Examines how communication is structured and managed in hybrid and distributed automotive project teams, including the challenges that affect information sharing, coordination, and stakeholder alignment.	Formal Communication Mechanisms, Informal Communication Practices, Information Transparency, Knowledge Sharing Across Functions
Theme 3: Cross-Functional Coordination and Role Alignment	Focuses on how roles, responsibilities, and dependencies are defined and coordinated across different functions, and how misalignment impacts project progress and outcomes.	Role Clarity, Decision Ownership, Dependency Management, Product Vision Alignment, Ownership and Accountability Across Functions
Theme 4: Team Dynamics and Conflict Management in Hybrid Environments	Captures the nature of team dynamics and the types of conflicts that arise in cross-functional and hybrid project environments, as well as strategies used to manage and resolve conflicts effectively.	Task-Related Conflict, Priority and Resource Conflicts, Interpersonal Tensions, Cultural and Professional Differences, Conflict Resolution Approaches, Impact of Unresolved Conflict on Performance

Theme 5: Managerial Practices for Enhancing Collaboration and Adaptability	Highlights managerial practices and leadership behaviors that support collaboration, adaptability, and resilience in automotive automation projects operating under conditions of uncertainty and technological complexity.	Effective Collaboration Practices, Leadership Skill Development, Communication Improvement Practices, Adaptability to Changing Requirements, Learning from Past Project Failures, Organizational Support for Collaboration, Managerial Recommendations for Hybrid Teams
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Table 3 Semi structured Interview Guide

Theme	Question No.	Interview Questions
Theme 1: Human-Centered Leadership in Cross-Functional Automotive Projects	Q1	Can you describe your role and your involvement in cross-functional automotive automation projects?
	Q2	How would you describe the leadership approach used in your current or most recent project?
	Q3	In what ways does leadership behavior influence trust, motivation, or collaboration within the team?
Theme 2: Communication Practices and Information Flow in Hybrid Teams	Q4	How communication is typically organized within your cross-functional project team?
	Q5	What communication challenges do you experience when working in hybrid or distributed environments?
Theme 3: Cross-Functional Coordination and Role Alignment	Q6	How are roles and responsibilities defined and coordinated across different functions in your project?
	Q7	Can you describe a situation where cross-functional misalignment affected project progress or outcomes?

Theme 4: Team Dynamics and Conflict Management in Hybrid Environments	Q8	What types of conflicts tend to arise in cross-functional or hybrid automotive teams?
	Q9	How conflicts are usually managed or resolved within your team?
Theme 5: Managerial Practices for Enhancing Collaboration and Adaptability	Q10	What practices or behaviors have been most effective in improving cross-functional collaboration in your projects?
	Q11	How do teams adapt to changes or uncertainty in automotive automation projects?
	Q12	Based on your experience, what recommendations would you give to product or project managers leading hybrid automotive teams?

4.1 Theme 1: Human-Centered Leadership in Cross-Functional Automotive Projects

This theme explores how human-centered leadership is practiced within cross-functional automotive automation projects and how such leadership influences trust, motivation, and collaboration among team members working in hybrid and multidisciplinary environments. The findings provide an overview of the leadership practices adopted in automotive automation teams and how these practices shape coordination, communication, and shared accountability across technical and non-technical functions.

The questions under this theme were designed to understand the nature of participants' roles in cross-functional automotive automation projects and to examine how leadership is enacted when multiple disciplines such as software, mechanical engineering, automation, quality assurance, system integration, and operations must work together. Detailed findings related to these questions are discussed below.

Q1: Can you describe your role and your involvement in cross-functional automotive automation projects?

Participants' Roles and Cross-Functional Involvement

Respondents described diverse roles within automotive automation projects, reflecting the cross-functional structure of modern automotive development environments. Participants represented areas including product and project management, software engineering, mechanical engineering, automation engineering, system integration, quality assurance, operations, and agile coaching. This variety provided a holistic understanding of how cross-functional teams operate and how leadership practices are experienced across different specialties.

Product and project managers specifically addressed their coordination and integrative roles, referring to the importance of strategic goals formulated and fulfilled on a technological level. A product manager expressed:

My role involves coordinating between software engineers, hardware teams, and business stakeholders to ensure product requirements are clearly translated into technical deliverables” *(Res1)*.

Similarly, a project manager described their leadership role as one of alignment and dependency management across functions:

I work closely with engineering, automation, and quality teams to align timelines and manage dependencies across functions” *(Res2)*.

These accounts suggest that leadership in automotive automation projects is less hierarchical and more facilitate, with an emphasis on orchestration, communication, and shared accountability.

“I am an automation engineer involved in designing and implementing automated systems for vehicle components. My work requires constant coordination with software developers and product managers to ensure system integration” *(Res3)*.

Similarly, software and mechanical engineers emphasized the importance of aligning software solutions with hardware limitations and project requirements. One participant stated:

“I collaborate with mechanical engineers and product owners to ensure the software aligns with hardware constraints and project goals” *(Res4)*.

A mechanical engineer also described frequent interaction with software and systems teams to ensure feasibility and smooth integration:

“I work as a mechanical engineer supporting automation projects. I interact regularly with software and systems teams to address design feasibility and ensure mechanical components integrate smoothly with automated systems” (*Res5*).

In addition, participants working in agile and leadership support roles described their involvement in strengthening collaboration and improving communication practices. An agile coach explained:

“As an Agile Coach, I support cross-functional automotive teams by facilitating collaboration, improving communication practices, and helping teams adapt agile principles in hybrid work environments” (*Res6*).

Leadership roles within engineering teams also reflected a people-centered approach, with an emphasis on supporting team members while coordinating tasks and resolving technical dependencies. One respondent noted:

“I lead a multidisciplinary engineering team responsible for automation modules. My role involves coordinating tasks across functions, resolving technical dependencies, and supporting team members in a hybrid setup” (*Res7*).

System integration specialists emphasized the need for collaboration across multiple functions to ensure compatibility between software and hardware components. One participant stated:

“I am involved in system-level integration for automotive automation projects. My role requires working closely with different functional teams to ensure compatibility and alignment across hardware and software components” (*Res8*).

Quality assurance respondents described their role as ensuring safety and performance compliance while collaborating closely with engineering and product teams to identify problems early. One participant explained:

“I work in quality assurance, ensuring that automated systems meet safety and performance standards. My involvement includes collaborating with engineering and product teams to identify issues early in the development cycle” (*Res9*).

Finally, one respondent highlighted a bridging role between development and manufacturing, showing how cross-functional coordination extends into production feasibility:

“My role bridges product development and manufacturing. I coordinate with automation engineers, product managers, and operations teams to ensure designs are feasible for large-scale production” (*Res10*).

Table 4 Respondents’ Roles and Involvement in Cross-Functional Automotive Automation Projects

Statement	Count
People management, team coordination, and cross-functional collaboration were the primary focus.	3
Technical execution and system level problem-solving formed the main focus, with teamwork required to support the integration process.	4
Emphasis was placed on enabling collaboration and synchronization between teams, without direct responsibility for people or project management.	3

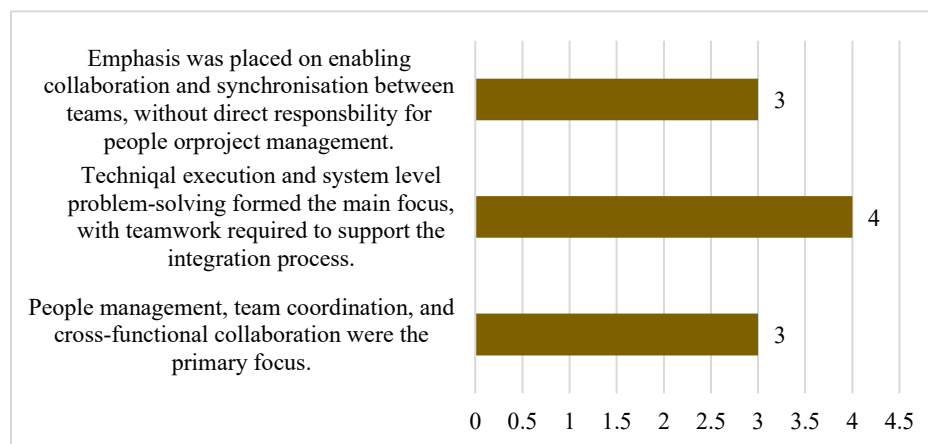


Figure 2 Graph representation of respondents’ Roles and Involvement in Cross-Functional Automotive Automation Projects

Q2: How would you describe the leadership approach used in your current or most recent project?

Leadership Approaches Practiced in Cross-Functional Automotive Automation Projects

Respondents described leadership approaches in their automotive automation projects as strongly human-centered, with an emphasis on collaboration, empowerment, trust-building, and shared ownership across cross-functional teams. Overall, participants indicated that leadership was less focused on command-and-control practices and more oriented toward enabling teamwork, facilitating communication, and supporting employees across hybrid and multidisciplinary work settings.

A key pattern across responses was the presence of collaborative and inclusive leadership. Participants reported that leaders encouraged open discussion and actively invited input from different functional areas, which strengthened trust and increased motivation. One participant explained:

“The leadership approach was largely collaborative and inclusive. Leaders encouraged discussions and inputs from various functions were, hence fostering trust among team members who were motivated to add value even beyond their own roles” (*Res1*).

Similarly, some respondents highlighted that leadership balanced empowerment with structure, where expectations were clearly communicated but team members were still given autonomy to contribute across functional boundaries. For example, one participant stated:

“The leadership aspect within my project was a balance between supportive leadership and structure, where there were set expectations, but also a certain degree of empowerment within teamwork, particularly between engineering and quality” (*Res2*).

Several participants described leadership as hands-on, particularly during critical project stages. This leadership approach was viewed as increasing leader accessibility, responsiveness, and team motivation. One respondent noted:

“A hands-on leadership style was promoted, particularly during critical stages. This made the leaders more accessible and responsive, resulting in improved communication and boosting the motivation level of the team members” (*Res3*).

In addition, servant leadership principles were strongly reflected in some accounts. Respondents described leaders as focusing on removing barriers and enabling team success rather than

supervising tasks. This was seen as strengthening collaboration between software and hardware teams. One participant explained:

“Our project was based on ‘servant leadership’ principles. Our leaders primarily worked on removing barriers rather than just supervising our work. The result was that our software and hardware teams fostered a collaborative environment” (*Res4*).

Another leadership aspect emphasized by participants was the importance of technical competence combined with mutual respect. Respondents indicated that when leaders understood engineering limitations and encouraged mutual understanding across disciplines, cooperation between cross-functional teams improved. One participant stated:

“The issue of leadership emphasized both technical competency and mutual respect. When there was mutual understanding of engineering limitations, this boosted mutual respect and cooperation between cross-functional teams” (*Res5*).

Agile-oriented leadership was also frequently mentioned, particularly in relation to hybrid and remote work contexts. Participants explained that adaptive leadership, transparency, and open feedback practices helped maintain trust and motivation even when teams were distributed. One respondent noted:

“Leadership style had been adaptive and congruent with the principles of Agile. They encouraged transparency and open feedback, which helped build trust and sustain motivation even for remote working” (*Res6*).

Respondents further described leadership as goal-oriented but people-centered. Regular check-ins, recognition of contributions, and attention to team morale were perceived as important leadership behaviors that strengthened motivation. One participant explained:

“The leadership was goal-oriented but people-centered. Team member check-ins and recognition of individual contributions made a positive impact on team Morale” (*Res7*).

A strong theme of shared accountability also emerged across responses. Participants reported that leaders encouraged cross-functional ownership rather than silo-based responsibility, ensuring that systems, software, and mechanical components were managed collectively. One respondent stated:

“Cross-functional ownership was encouraged over ownership by silos. Such practices ensured there was shared accountability related to systems, software, and mechanics” (*Res8*).

Quality-focused participants also emphasized that leadership behaviors strongly influenced collaboration outcomes. When leaders involved quality teams early in decision-making, trust increased and misunderstandings were reduced. One respondent noted:

"Leadership behavior was important in collaboration too. When leaders actively involved in quality teams who received early in decision-making, it increased trust and minimizing misunderstanding and conflict" (*Res9*).

Finally, respondents highlighted pragmatic leadership practices that supported alignment between development and manufacturing. This alignment was seen as essential for collaboration and ensuring stakeholder integration across the project lifecycle. One participant explained:

“It is important to mention that the leadership style adopted was pragmatic. Leaders enabled alignment between development and manufacturing teams, which ensured collaboration between project stakeholders” (*Res10*).

Table 5 Summary of Leadership Approaches Practiced in Cross-Functional Hybrid Automotive Projects

Statement	Count
Leadership focused on trust, motivation, collaboration, and recognition of individual contributions	5
Leadership emphasized accessibility, responsiveness, technical alignment, and goal achievement.	2
Leadership focused on enabling transparency, shared ownership, and alignment across teams without direct control or technical enforcement.	3

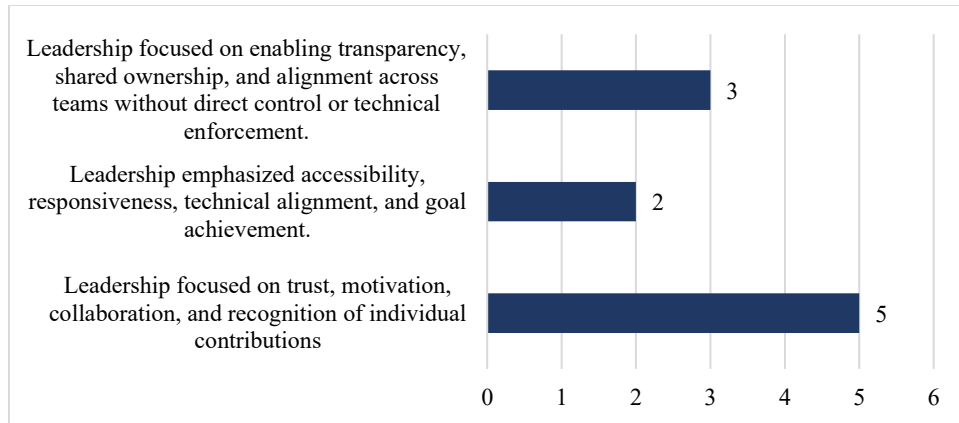


Figure 3 Graph representation of Summary of Leadership Approaches Practiced in Cross-Functional Hybrid Automotive Projects

Q3: In what ways does leadership behavior influence trust, motivation, or collaboration with in the team?

Influence of Leadership Behavior on Trust, Motivation, and Collaboration

Respondents consistently emphasized that leadership behavior plays a critical role in shaping trust, motivation, and collaboration within cross-functional automotive automation projects. Participants explained that leadership influences not only technical coordination, but also the interpersonal dynamics that determine how effectively teams cooperate across functions such as software, hardware, systems integration, quality, and production. Overall, leadership behaviors were viewed as central in promoting transparency, shared accountability, team confidence, and consistent collaboration in both hybrid and multidisciplinary environments.

A prominent influence highlighted by participants was collaborative leadership through inclusive decision-making. Respondents noted that when leaders actively involved cross-functional members in decisions, trust increased and team members felt valued, regardless of whether they came from technical or non-technical functions. One participant explained:

“The leadership approach was collaborative, with an emphasis on involving cross-functional teams in decision-making processes. Leaders encouraged open dialogue and valued contributions from both technical and non-technical stakeholders” (*Res1*).

Another major leadership contribution related to establishing structure, clarity, and shared goals. Participants described that when leaders set clear objectives from the beginning while still allowing flexibility in execution, it improved accountability and efficiency. One respondent stated:

“The leadership in the project was organized and had clear goals of what they intended to achieve. There were set objectives right from the start, with a degree of freedom in the way they were carried out that assisted efficiency and accountability” *(Res2)*.

Several respondents described participative leadership as an important factor in maintaining collaboration and direction, especially during complex integration tasks. Frequent interactions with leaders were seen as motivating and supportive, ensuring that teams stayed aligned while navigating technical challenges. One participant noted:

“The leadership style was largely participative. The team members had frequent interactions with the leaders, and proper direction was given, especially in complex integration activities” *(Res3)*.

Supportive leadership was also identified as a strong driver of collaboration. Participants explained that leaders who focused on enabling teams rather than micromanaging them strengthened alignment between software and hardware groups. One respondent explained:

“Our project adopted a supportive leadership approach where leaders concentrated on supporting teams rather than micromanaging them. This helped to ensure alignment between software and hardware teams” *(Res4)*.

Technical credibility and pragmatic decision-making were further described as leadership behaviors that strengthened trust. Respondents indicated that when leaders demonstrated awareness of engineering constraints and challenges, team members gained confidence in leadership decisions. One participant stated:

“The leadership was technically informed and pragmatic. The leaders showed sound awareness of the engineering challenges that proved beneficial while making informed decisions and gaining the confidence of the team members” *(Res5)*.

Participants also noted that adaptive leadership aligned with agile principles enhanced motivation and responsiveness. Leaders who adjusted their engagement levels depending on project needs were seen as supporting both flexibility and team effectiveness. One respondent explained:

“A strong leadership approach that adapted to fit the agile mindset. Their level of leadership engagement varied according to the needs of the project” *(Res6)*.

In addition, people-centered leadership behaviors were strongly associated with motivation and consistency across functions. Participants described that leaders who emphasized team wellness, regular communication, and clear expectations supported both performance and stable collaboration. One respondent noted:

“It was people-centric, with a strong focus on team wellness and performance. Regular communication and clarity of expectations ensured consistency between functions” *(Res7)*.

Cross-functional ownership was also identified as a key leadership behavior that reduced siloed working patterns. Participants emphasized that when leaders encouraged teams to collaborate beyond their own functional boundaries, it strengthened teamwork and reduced fragmentation. One participant explained:

"The leadership style that was promoted was cross-functional ownership. Collaboration beyond functions was taken into account, which helped to minimize silos"*(Res8)*.

Leadership behaviors were also linked to proactive risk management and quality collaboration. Respondents noted that leaders who emphasized early identification of risks and quality issues helped teams address problems before they escalated, supporting smoother cooperation. One participant stated:

“There was proactive leadership action, especially with regards to risk and quality. It was easier to address problems earlier” *(Res9)*.

Finally, respondents highlighted inclusive and enabling leadership behaviors as important for overall project alignment. Participants explained that leaders who acted as coordinators across development, operations, and production strengthened trust and collaboration among stakeholders across the project lifecycle. One respondent stated:

“The leadership approach was very inclusive and enabling in nature. The leaders played the role of project coordinators between development, operational, and production teams to facilitate overall project alignment” *(Res10)*.

Table 6 Leadership Behaviors Influencing Trust, Motivation, and Collaboration in Hybrid Teams

Statement	Count
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Leadership behavior strengthened trust and collaboration through open communication, team support, and inclusive involvement.	5
Leadership behavior enhanced confidence and accountability by providing clarity, technical guidance, and early problem resolution.	3
Leadership behavior enabled coordination and reduced silos by adjusting involvement and encouraging shared responsibility across teams.	2

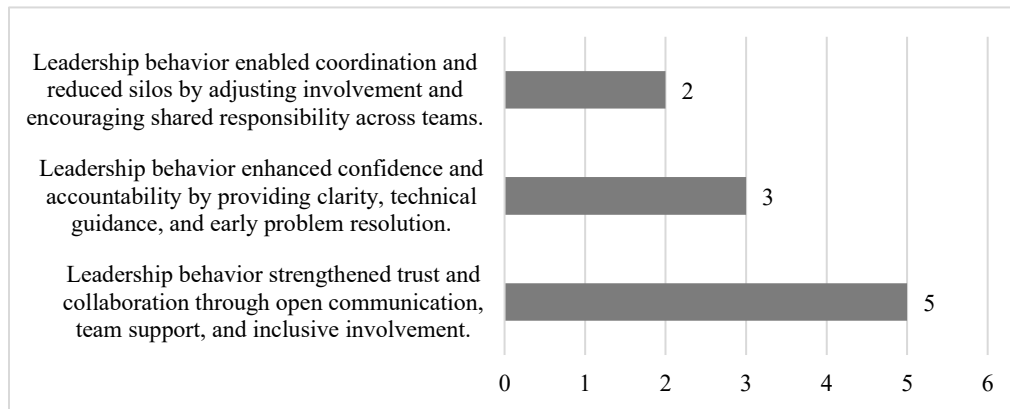


Figure 4 Graph representation of Leadership Behaviors Influencing Trust, Motivation, and Collaboration in Hybrid Teams

4.2 Theme 2: Communication Practices and Information Flow in Hybrid Teams

This theme examines how communication is structured and maintained within cross-functional automotive automation teams operating in hybrid or distributed work environments. The findings highlight the formal and informal communication channels used by participants to ensure coordination and effective information flow across functions. In addition, this theme evaluates how digital collaboration tools support teamwork and identifies key communication challenges that emerge when teams are not physically co-located.

The questions under this theme were designed to explore how communication practices influence coordination, clarity, and responsiveness in complex automotive automation projects. The findings also aimed to reveal common barriers affecting information exchange, decision-making speed, and collaboration quality in hybrid project settings. Detailed findings related to these questions are discussed below.

Q4. How communication typically organized within your cross-functional project team?

Communication Structures and Information Flow in Hybrid Cross-Functional Teams

Respondents reported that communication in hybrid automotive automation projects is primarily organized through structured agile routines, scheduled coordination meetings, and the use of digital collaboration tools. Participants frequently mentioned formal communication mechanisms such as daily stand-ups, sprint meetings, milestone reviews, cross-functional design sessions, and system integration meetings. Digital platforms such as Microsoft Teams, Jira, dashboards, and project management tools were consistently described as central to maintaining information flow across functions.

Several participants highlighted that agile ceremonies form the backbone of communication and coordination. For example, one respondent noted:

“Communication has been organized through regular sprint meetings, daily stand-ups, and digital platforms like Teams and Jira. But in a hybrid model, lag in responses, less opportunistic communication, could create a misinterpretation between business/tech teams sometimes” (*Res1*).

Similarly, respondents described weekly planning meetings as important for cross-functional alignment, particularly when multiple teams must coordinate tasks and dependencies. However, participants also emphasized that hybrid and distributed work introduce scheduling constraints, especially across time zones. One participant explained:

“We use planning meetings on a weekly basis to coordinate activities across the various functions. Getting people’s schedules aligned across time zones poses a challenge, particularly if urgent matters require fast decision making” (*Res2*).

Participants also reported that most communication takes place through online meetings and digital platforms. While this ensures that team members remain updated, respondents noted that complex technical discussions are more difficult to conduct virtually compared to face-to-face interactions. One respondent stated:

“Most of their communications are carried out through online meetings and project management. Although everybody is kept updated, there are difficulties in dealing with complex technological talks online compared to offline communications” (*Res3*).

Some participants explained that communication is organized through designated channels based on functional topics such as software and hardware. This approach was seen as supporting structured information exchange; however, the lack of immediate feedback in hybrid environments sometimes led to misunderstanding of requirements. One respondent noted:

“We have designated communication channels for different topics such as software and hardware. With the hybrid model that we have at Tech Whirl, sometimes the absence of instant feedback results in misconceptions about requirements” (*Res4*).

In addition, respondents described communication being structured around milestone reviews and cross-functional design meetings, particularly when projects involve global teams and complex deliverables. One participant explained:

“Communication is organized around milestone reviews and cross-functional design meetings” (*Res5*).

Agile-oriented participants further emphasized that transparency is a key communication principle in hybrid teams, but it requires additional effort because not all members participate equally in online meetings. One respondent stated:

“Aside from Agile activities such as standups, retrospectives, and sprint review, we make sure to have transparency. It requires more effort to have open communication in hybrid teams because some of the team members will not contribute during online meetings” (*Res6*).

Other respondents explained that information often flows through team leads who coordinate updates through digital tools and sync-ups. However, participants noted that important updates may be missed when teams rely on multiple communication channels simultaneously. One participant explained:

“Information flows through team leads who coordinate through online tools and sync-ups. One of the challenges is making sure that everybody hears about important changes, because messages can get lost through different communication channels” (*Res7*).

Some respondents described communication as highly formalized, particularly in system integration contexts where all functional teams must align. However, participants noted that

distributed teams may experience communication friction when expectations and communication styles differ across locations. One participant stated:

“Communication is much formalized, and system integration meetings are where all the functions are brought together. There are potential issues in distributed environments if communication styles and expectations are different. This would create communication friction” (*Res8*).

Quality-focused respondents highlighted that communication is supported through quality review processes and early involvement in project discussions. However, participants reported that distributed work slows feedback cycles and reduces the speed of problem-solving. One respondent explained:

“We facilitate communication via quality review and early engagement with project conversations. The biggest challenge is reacting to problems quickly in distributed teams, which slows down feedback cycles” (*Res9*).

Finally, respondents described the use of cross-functional forums and dashboards to integrate communication and monitor progress. However, participants noted that hybrid environments may make it more difficult to build trust, which can reduce openness and honesty when discussing concerns. One participant stated:

“Our communication is integrated using cross-functional forums and dashboards to monitor progress. In a hybrid environment, trust is more difficult to establish, and this may make it difficult to be frank when discussing concerns” (*Res10*).

Table 7 Communication Organization and Information Flow Practices in Hybrid Cross-Functional Teams

Statement	Count
Communication is structured through meetings, agile rituals, and formal channels	4
Digital tools enable information flow but dominate communication in hybrid teams	3
Hybrid work creates challenges like delays, misinterpretation, weak feedback, and time-zone issues	1
Trust and open participation are harder to build in hybrid communication	2

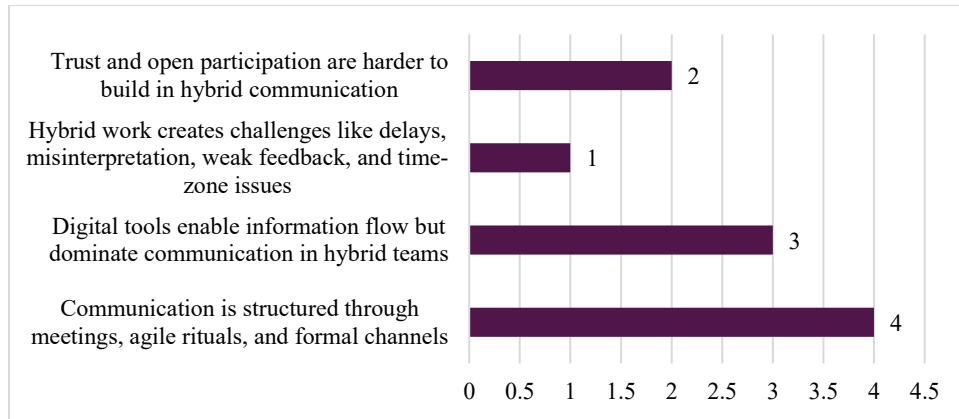


Figure 5 Graph representation of Communication Challenges Experienced in Hybrid and Distributed Project Environments

Q5. What communication challenges do you experience when working in hybrid or distributed environments?

Communication Challenges in Hybrid and Distributed Automotive Automation Teams

Respondents identified several communication challenges that affect coordination, clarity, responsiveness, and collaboration when working in hybrid or distributed automotive automation environments. Overall, participants emphasized that the lack of physical co-location reduces informal interaction, slows feedback cycles, and increases the likelihood of misunderstanding, particularly in complex technical projects where rapid clarification and shared context are essential.

A commonly reported challenge was the absence of informal communication and spontaneous discussions. Participants noted that when teams are remote, opportunities for quick clarification and casual problem-solving decrease, which can cause misunderstandings to persist longer and require additional formal meetings. One respondent explained:

“A key issue might be the absence of informally communicating. Without spontaneous discussions, misunderstandings can persist longer and require additional meetings to resolve”

(Res1)

Time zone differences were also highlighted as a major barrier, especially in global projects requiring urgent decisions during critical phases. Respondents explained that delays in feedback and decision-making are more frequent when teams are distributed across different working hours. One participant stated:

"Coordinating across time zones is challenging. Sometimes it causes delays in feedback and decision-making processes regarding critical phases of projects when urgent situations occur" (*Res2*).

Several participants emphasized that technical communication becomes less effective in remote settings. Respondents noted that complex system discussions, particularly those involving system interactions and integration, are harder to conduct without physical tools such as whiteboards and face-to-face engagement. One respondent explained:

"Technical talk becomes less effective at a distance. It becomes harder to describe system interactions with physical whiteboards and actually seeing each other" (*Res3*).

Slow response times in online communication were also reported as a frequent challenge. Participants described that because team members are not always online simultaneously, even simple questions may take longer to resolve, slowing overall project progress. One participant noted:

"Response times in online communication may slow progress. Team members may not be online at the same time. It takes a longer time for a simple question to be clarified" (*Res4*).

Participants further highlighted that hybrid work can reduce the sense of inclusion and participation for members who are not physically present in meetings. Respondents explained that this imbalance may lead to reduced engagement, weaker cooperation, and less effective problem-solving. One respondent stated:

"The sense of participation for people not physically present could be lessened compared to people in the meeting. This could result in less participation during discussion and less cooperation during problem-solving" (*Res5*).

Similarly, respondents noted that ensuring equal participation in online meetings is difficult, as some team members hesitate to speak up. This challenge was perceived as limiting idea-sharing and reducing diversity of perspectives. One participant explained:

"Ensuring equal participation in online meetings is not easy. Some team members are reluctant to speak up, which restricts diverse ideas from coming up" (*Res6*).

Another major challenge reported was fragmented information flow across multiple digital tools. Participants stated that when communication occurs through emails, chat discussions, project sites, and other platforms simultaneously, important updates can be missed. One respondent explained:

"Information might be fragmented over several digital tools. Critical updates could be overlooked when communication takes place through emails, discussions, and project sites" (*Res7*).

Cultural differences and varying communication styles were also described as more visible in distributed environments. Participants noted that misunderstandings or tensions may occur when expectations about communication differ across teams. One respondent stated:

"Differences in communication style and cultural expectations become more apparent in distributed teams, and sometimes there can be miscommunications or tension within them" (*Res8*).

Quality assurance and feedback processes were also perceived as more difficult when teams are not physically together. Respondents explained that giving feedback and resolving quality issues may take longer when problems cannot be directly observed or checked in person. One participant noted:

"It's more challenging to give feedback when the team isn't all in the same room. Problems with quality can take longer to fix because it can't be easily checked in person" (*Res9*).

Finally, respondents emphasized that trust and relationship-building are harder to develop in remote work environments. Participants explained that when team members do not meet regularly, they may be less willing to openly share concerns or difficulties, which reduces transparency and collaboration. One respondent stated:

"The wider context "Trust and relationship-building can be harder to achieve when working remotely. Also, when team members don't meet on a regular basis, they may not feel free to share concerns or difficulties with each other" (*Res10*).

Table 8 Communication Challenges Experienced in Hybrid and Distributed Project Environments

Statement	Count
Delays from time zones and slow responses make communication and decisions harder	3
Lack of informal interaction and difficulty building trust reduce openness	2

Lower participation and engagement of remote members in discussions	2
Difficulty handling complex technical discussions without face-to-face interaction	1
Fragmented tools and cultural differences cause misunderstandings	2

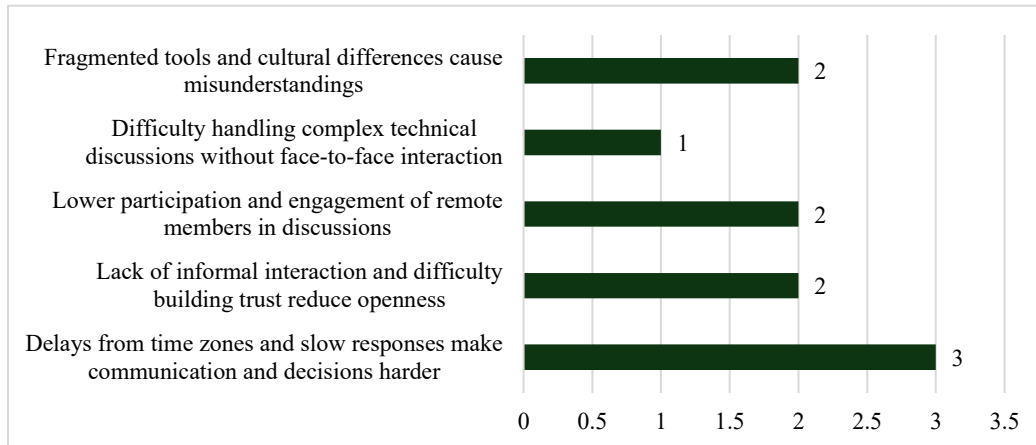


Figure 6 Graph representation of Communication Challenges Experienced in Hybrid and Distributed Project Environments

4.3 Theme 3: Cross-Functional Coordination and Role Alignment

This theme explores how roles and responsibilities are defined, shared, and coordinated across the different functional units involved in automotive automation projects. The findings provide insight into the level of role clarity within cross-functional teams and the coordination mechanisms used to manage interdependencies between engineering, software, quality, system integration, management, and manufacturing functions.

The questions under this theme were designed to understand how role alignment (or lack of alignment) influences project progress, decision-making, accountability, and overall project outcomes in hybrid work environments. In particular, the findings highlight how unclear ownership, conflicting priorities, and communication gaps can lead to delays, rework, and integration failures. Detailed findings related to these questions are discussed below.

Q6. How are roles and responsibilities defined and coordinated across different functions in your project?

Role Definition, Coordination Mechanisms, and Alignment across Functions

Respondents described that roles and responsibilities in cross-functional automotive automation projects are typically defined during project initiation and formal planning stages. Participants reported the use of structured mechanisms such as project kick-off sessions, responsibility matrices, design documentation, project plans, agile role definitions, and system-level mapping. Coordination was commonly maintained through cross-functional reviews, weekly coordination calls, sprint planning meetings, governance forums, milestone checkpoints, and Agile ceremonies supported by visual boards and tracking tools.

Several participants emphasized the use of formal responsibility definition methods such as responsibility matrices and kick-off alignment processes. However, respondents noted that despite such structures, role ambiguity can still occur, particularly in system integration activities where software and hardware responsibilities overlap. One participant explained:

“Roles are determined in the project kick-off process using responsibility matrices. Coordination occurs through cross-functional reviews. This project experienced a several-week delay in system integration because of a lack of clarity on whether software development or hardware engineering owned the issue” (*Res1*).

Similarly, respondents reported that project plans often define functional scopes clearly, while coordination across dependencies is managed through regular cross-functional calls. However, participants indicated that alignment may be lost when key information is not shared across functions, leading to rework. One respondent noted:

“Each function has been identified with well-defined scopes in the project plan, and dependencies are coordinated on weekly coordination calls. Alignment was lost once when engineering change inputs were not shared with quality teams, causing rework” (*Res2*).

Participants also explained that responsibilities are frequently assigned based on technical expertise, with integration points managed through joint discussions. However, respondents highlighted that unclear ownership of interface responsibilities can lead to duplicated work or overlap between teams. One participant stated:

“Responsibilities are assigned according to technical knowledge, with integration points being discussed in joint meetings. Lack of understanding about interface responsibility led, in the past, to overlap between teams” (*Res3*).

Agile-based role structures were also frequently mentioned. Participants explained that roles such as product owners, technical leads, and sprint-based task ownership help structure responsibility distribution. However, respondents noted that conflicts between functional priorities may still occur, particularly between software and mechanical teams, resulting in delivery delays. One respondent explained:

“For role definitions, we do Agile roles, supported by product owners and tech leads. For instance, during a particular sprint, priorities clashed between software and mech groups, resulting in latency on the delivery of the features” (*Res4*).

In addition, some participants reported that design documents play a major role in defining responsibilities and are reviewed during milestone assessments. However, respondents highlighted that when design constraints are not communicated early, misalignment may lead to late-stage changes, particularly in mechanical components. One participant stated:

"Roles are specified by design documents, which are assessed at each milestone. Misalignment occurred due to a lack of communication about design constraints earlier, causing design changes to mechanical components towards the end” (*Res5*).

Agile coaching perspectives also emphasized that synchronization is supported through ceremonies and team visual boards. Nevertheless, respondents noted that lack of shared understanding of priorities can still affect sprint performance and alignment. One respondent explained:

“In Agile Coaching, I see how tasks are synchronized through ceremonies and team visual boards. However, there could be a lack of alignment when teams don't align in understanding priorities, impacting the sprint objectives” (*Res6*).

Several participants highlighted the role of team leads in coordinating ownership and responsibility across functions. However, respondents indicated that decision ownership can become unclear during critical phases, resulting in delays in approvals and project progress. One participant noted:

“Team leads provide task ownership coordination between functions during planning meetings. There was confusion over decision ownership during one phase of a project, which led to delays in approvals” (*Res7*).

System integration respondents described that system-level responsibilities are often mapped across teams using integration checkpoints. However, participants explained that integration failures may occur when system-level requirements are not consistently aligned across teams. One respondent stated:

"System-level responsibilities are mapped across teams with integration checkpoints. There are integration failures due to the lack of alignment of the system-level requirements in the past" **(Res8)**.

Quality assurance participants emphasized that quality roles are most effective when quality teams are involved early in development. Respondents noted that when quality engagement is delayed, defects may be identified too late, negatively affecting outcomes. One participant explained:

"Quality roles are defined so that early involvement is made in development. In a situation where quality was brought into a certain project late, certain defects that had resulted were identified late" **(Res9)**.

Finally, respondents reported that coordination is sometimes managed through cross-functional governance meetings and roadmaps, particularly when aligning development and manufacturing. However, participants described that misalignment between these functions can result in designs that are not production-ready, causing delays. One respondent stated:

"Coordination is handled by cross-functional governance meetings as well as roadmaps. Once, there was a misalignment between development teams and manufacturing teams that resulted in designs not ready for manufacturing, hence delays" **(Res10)**.

Table 9 Summary of Role Definition and Coordination across Functions

Simple Statement	Count
Roles are defined using kick-off planning and responsibility tools.	2
Coordination is managed through regular cross-functional meetings.	2
Agile roles and ceremonies are used to organize responsibilities.	2
Role clarity is supported through design documents and milestone reviews.	1

Integration checkpoints are used to align system-level responsibilities.	1
Early quality involvement improves alignment and reduces defects.	1
Unclear ownership leads to delays and rework.	1

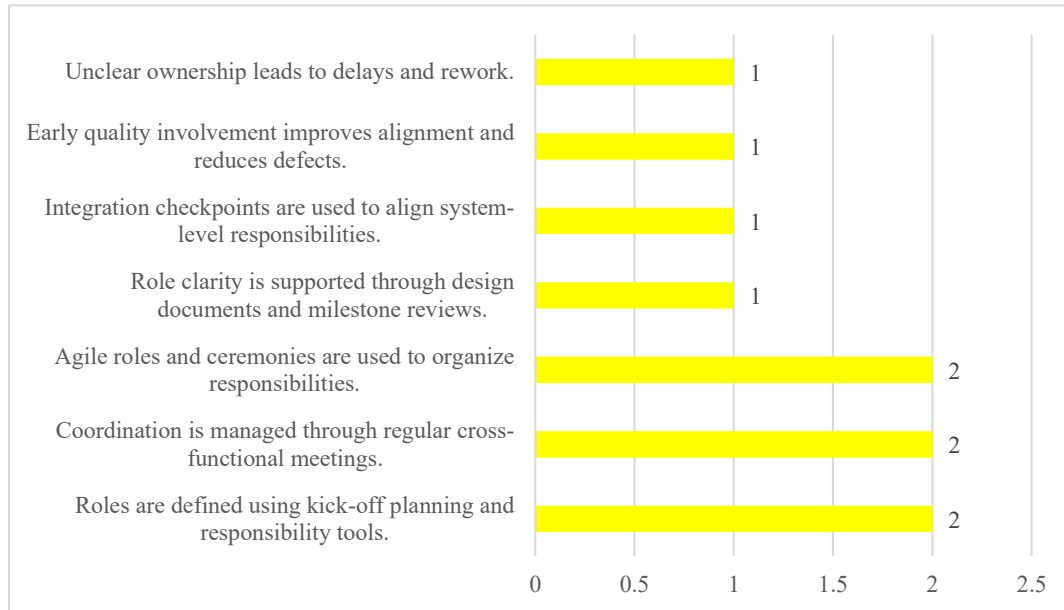


Figure 7 Graph representation of Summary of Role Definition and Coordination across Functions

Q7. Can you describe a situation where cross-functional misalignment affected project progress or outcomes?

Examples of Cross-Functional Misalignment and Its Impact on Project Outcomes

Respondents provided multiple examples showing that cross-functional misalignment can significantly affect project progress, integration success, delivery timelines, cost, and team morale in automotive automation projects. Participants explained that misalignment often emerges when information is not shared in time, when teams interpret requirements differently, when responsibilities overlap, or when decision-making authority is unclear. Overall, these issues were reported to cause rework, failed integration testing, delayed schedules, and reduced confidence across teams operating in hybrid environments.

A frequently reported misalignment issue involved late communication of requirement changes between functions. Participants described situations where software requirements changed but were not communicated to hardware teams in time, resulting in compatibility failures during integration and causing major project delays. One respondent explained:

“There was an instance where the software requirements had changed, but the hardware team had not been notified in time. It led to compatibility problems during integration, which caused a delay in the project by several weeks” (*Res1*).

Similarly, misalignment between engineering and quality teams was reported as causing additional testing cycles and delays. Respondents explained that when design changes are made without involving quality teams, projects experience increased rework and schedule disruptions. One participant noted:

"One phase involved engineering, which made design changes that were unaligned with the quality team. This caused more testing cycles, which resulted in a delayed delivery schedule" (*Res2*).

Several participants described misalignment arising from unclear ownership boundaries, particularly when multiple teams assumed responsibility for the same system interface. This overlap led to duplicated effort and confusion about responsibilities. One respondent stated:

“The misalignment happened where two teams took ownership of the same interface. This resulted in repetitions of efforts and misunderstandings about who should do what” (*Res3*).

Other respondents described how competing functional priorities can disrupt sprint objectives and delay delivery. Participants explained that software and mechanical teams sometimes focus on different priorities, resulting in reduced alignment on sprint outcomes. One participant explained:

“There were different priorities for both the software and mechanical teams, and this caused a sprint objective to be violated. While one team worked on performance optimization, the other engaged in addressing design constraints” (*Res4*).

Late communication of constraints between teams was also reported as a major contributor to last-minute changes. Participants described cases where mechanical limitations were communicated too late to software teams, forcing late-stage modifications close to release deadlines. One respondent stated:

“There was a point where information on mechanical limitations had to be relayed too late to the software side. This caused last-minute changes near the deadline for the release” (*Res5*).

Agile coaching perspectives highlighted that misalignment can also occur when teams interpret the product backlog differently. Respondents explained that when teams do not share a common understanding of backlog priorities and user expectations, features may be developed that fail to meet intended outcomes. One participant noted:

“It is where I saw the misalignment when teams understood the product backlog in different ways. There was a situation where features developed did not meet user expectations” (*Res6*).

Participants also emphasized that unclear decision-making authority can lead to extended discussions and deferred decisions, particularly during critical integration phases. This misalignment slows progress and delays project execution. One respondent explained:

“Unclear decision authority has once led to extensive discussions between functions about which decisions needed to be made. This meant that key decisions were deferred, hence delaying the process during the critical phase of integrating the functions” (*Res7*).

System-level misalignment was also highlighted as a significant issue. Respondents reported that when different functional groups interpret system requirements inconsistently, integration testing can fail, resulting in more meetings, additional coordination efforts, and extended project schedules. One participant stated:

“System requirements weren't met consistently by different groups, and so integration during the testing stage failed. As a result, more coordination meetings were needed, increasing the overall project schedule” (*Res8*).

Quality assurance misalignment was further described as a major cause of late defect discovery. Participants noted that when quality teams are excluded from early design meetings, defects are identified late in the process, negatively affecting project timing and reducing team morale. One respondent explained:

“The project did not include quality assurance in initial design meetings. Many defects were discovered late. This has harmed project timing as well as staff morale” (*Res9*).

Finally, respondents described misalignment between development and manufacturing teams as affecting feasibility and commercialization readiness. Participants explained that when designs are

not aligned with manufacturing constraints, product redesign becomes necessary, resulting in delays and increased costs. One participant stated:

"The development team was not aligned with the manufacturing team on product feasibility. The product design required changes, which resulted in delays and expenses even before commercialization" (*Res10*).

Table 10 Responses on Cross-Functional Misalignment and Its Impact

Statement	Count
Late or missing communication between functions caused delays and rework.	3
Misalignment led to integration failures and schedule delays.	2
Lack of early quality involvement caused defects to be found late.	1
Two teams owned the same interface, causing duplication and confusion.	1
Different priorities between software and mechanical teams caused sprint failure.	1
Unclear decision authority delayed approvals and progress.	1
Development and manufacturing were misaligned, causing redesign and extra cost.	1

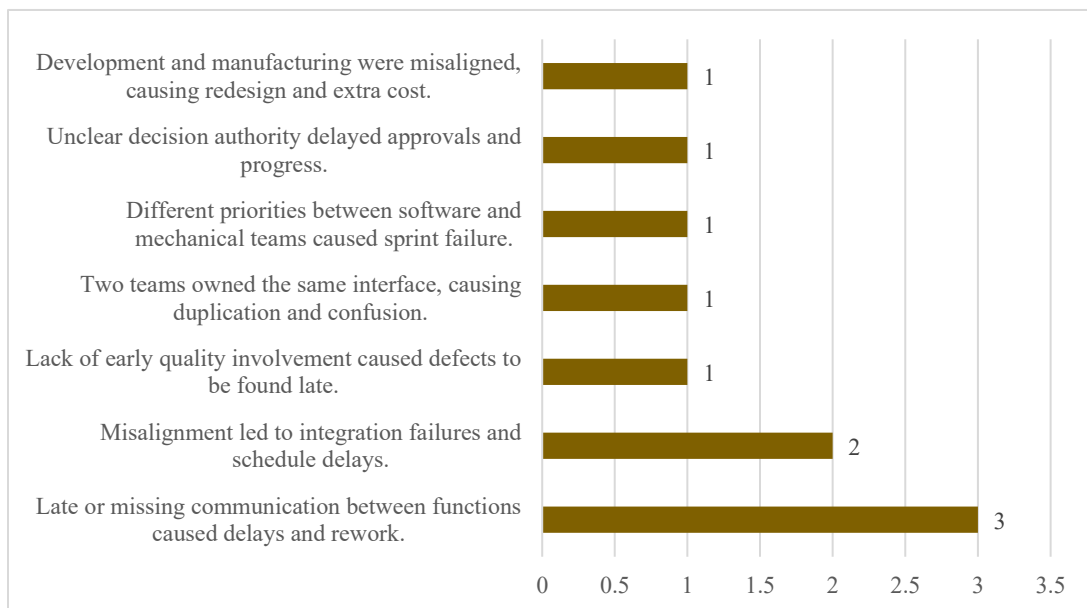


Figure 8 Graph representation of Responses on Cross-Functional Misalignment and Its Impact

4.4 Theme 4: Team Dynamics and Conflict Management in Hybrid Environments

This theme explores the nature of team dynamics and the types of conflicts that commonly arise in cross-functional automotive automation projects operating in hybrid or distributed environments. The findings provide insight into interpersonal, task-related, and process-based conflicts, as well as how hybrid work arrangements influence trust, collaboration, and working relationships among team members.

The questions under this theme were designed to examine the main sources of conflict in hybrid cross-functional teams and to understand how such conflicts affect team performance, coordination, and overall project success. The findings also highlight how differences in priorities, work styles, communication practices, and role expectations can create tension, particularly when teams are geographically dispersed. Detailed findings related to these questions are discussed below:

Q8. What types of conflicts tend to arise in cross-functional or hybrid teams?

Types of Conflict in Cross-Functional and Hybrid Automotive Automation Teams

Respondents reported that conflicts in hybrid and cross-functional automotive automation teams arise from a combination of task-related pressures, interpersonal differences, process misunderstandings, and coordination challenges. Participants emphasized that the complexity of automotive automation projects, combined with hybrid working conditions, often increases the likelihood of conflict, especially when communication is delayed, priorities differ across functions, or decision-making authority is unclear.

A frequently reported conflict type involved task-related disagreements, particularly when functional priorities differ between teams such as engineering and product management. Participants noted that these conflicts become more intense when deadlines are tight and teams must make trade-offs between delivery speed, technical quality, and product requirements. One respondent stated:

"Conflicts related to the task often arise when priorities differ from the engineering team to the product team, especially when deadlines are tight" (*Res1*).

Interpersonal conflicts were also described as common, especially when team members have different working styles, communication preferences, or expectations about collaboration. Respondents explained that such interpersonal tensions can affect trust and cooperation, particularly in hybrid environments where misunderstandings are harder to resolve quickly. One participant noted:

"Interpersonal conflicts arise sometimes when there are differences in working styles and communication preferences among team members" (*Res2*).

Process-related conflicts were another recurring issue. Participants reported that conflicts may arise when team members interpret Agile practices, workflows, or project procedures differently, leading to disagreement about how tasks should be executed and coordinated. One respondent explained:

"Process conflicts arise in teams when interpretations of agile processes or project workflows differ" (*Res3*).

Several respondents highlighted technical conflicts between software and mechanical teams, often caused by incomplete understanding of constraints across functions. Participants noted that when teams do not fully understand each other's technical limitations, disagreements may emerge over feasibility, timelines, and design decisions. One participant stated:

"Conflicts between software and mechanical teams arise when technical constraints are not fully understood across functions" (*Res4*).

Hybrid work was also described as contributing to coordination conflicts. Respondents explained that informal discussions often occur among co-located members, and remote participants may be unintentionally excluded. This can lead to misunderstandings and feelings of disengagement. One respondent noted:

But sometimes hybrid teams experience the complication of coordination conflicts because some remote members are bypassed in informal discussions, leading to misunderstandings" (*Res5*).

Resource-related conflicts were also identified as a challenge in cross-functional environments. Participants explained that when multiple teams compete for the same specialized technical skills, tools, or equipment, tension may arise, especially under tight schedules. One respondent stated:

Conflicts over resources arise when several teams require the same technical ability or equipment at the same time” (*Res6*).

Decision authority conflicts were described as another significant issue, particularly in cross-functional projects where approval responsibilities may not be clearly assigned. Respondents noted that ambiguity about who has final decision-making power can create delays and frustration across teams. One participant explained:

"Conflicts on decision authority arise when there is ambiguity among the team members as to where the authority for final approval on design or implementation decisions lies” (*Res7*).

Participants also reported timing conflicts as common in distributed teams, especially when teams operate across different time zones. Respondents explained that scheduling critical meetings and aligning milestone deadlines becomes more difficult, increasing stress and coordination problems. One participant stated:

Timing conflicts create problems across time zones, especially where critical meetings or milestone deadlines have to be set” (*Res8*).

Cultural differences were also identified as a cause of interpersonal conflict in distributed teams. Participants noted that differences in cultural norms and communication styles can undermine trust and create misunderstandings that affect collaboration. One respondent explained:

“Interpersonal conflicts in distributed teams are also prevalent due to cultural differences, which can undermine team trust and collaboration” (*Res9*).

Finally, respondents highlighted that conflicts may arise due to differences in expectations around documentation and reporting standards. Participants explained that teams may disagree about how much documentation is necessary and what reporting format should be followed, leading to friction between functions. One participant stated:

“Conflicts may occur because of expectations about different standards for documentation and reporting between teams” (*Res10*).

Table 11 Conflict Types and Conflict Resolution Practices in Cross-Functional Hybrid Teams

Statement	Count
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Roles and responsibilities are formally defined using plans, matrices, Agile roles, or design documents	4
Coordination is managed through cross-functional meetings, reviews, Agile ceremonies, and governance forums	2
Misalignment of priorities across functions affected delivery timelines	2
Late or weak involvement of certain functions (e.g., quality, manufacturing) led to defects or delays	2

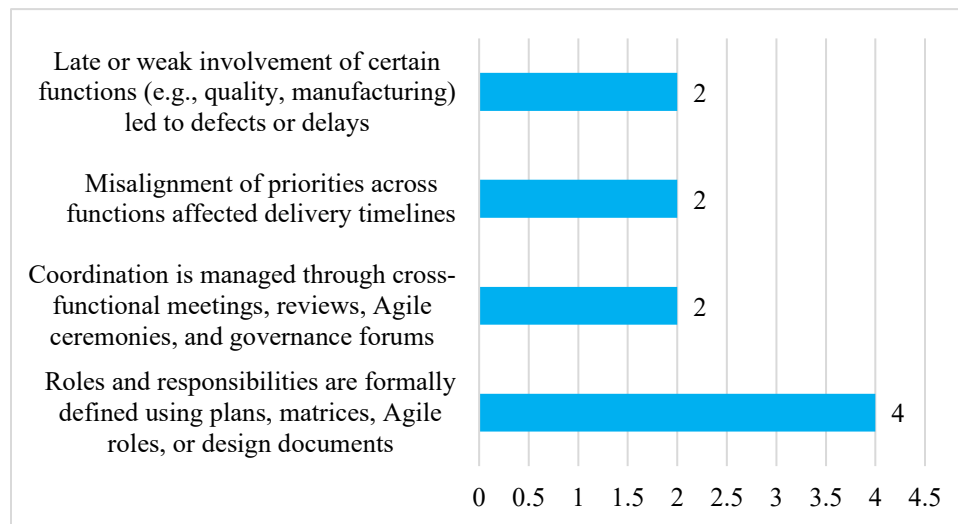


Figure 9 Graph representation of Conflict Types and Conflict Resolution Practices in Cross-Functional Hybrid Teams

Q9. How conflicts are usually managed or resolved within your team?

Conflict Management and Resolution Strategies in Hybrid Cross-Functional Teams

Respondents explained that conflicts in cross-functional and hybrid automotive automation teams are generally managed through structured communication, collaborative problem-solving, and leadership facilitation. Overall, participants indicated that conflict resolution is most often handled at the team level through open discussion, agile routines, and coordination meetings, while escalation to senior leadership occurs only when conflicts cannot be resolved internally.

A common approach described by respondents was resolving disputes through regular coordination meetings, where issues are openly discussed and solutions are jointly developed.

Participants emphasized that transparency and shared discussion help prevent conflicts from escalating. One respondent stated:

"Most disputes are dealt with in coordination meetings held each week. Matters are brought out in the open as solutions are pursued jointly" (*Res1*).

Leadership support was also described as important, particularly for interpersonal conflicts. Respondents explained that team leaders often act as mediators by encouraging communication and ensuring that all perspectives are heard. One participant noted:

"The team leaders may also help resolve interpersonal disputes among the team members by promoting communication and making sure all parties' ideas are shared" (*Res2*).

Process-related conflicts were reported to be managed by referring back to established workflows, Scrum events, and agreed project rules. Participants highlighted that structured frameworks help reduce confusion and bring teams back into alignment. One respondent explained:

"Process conflicts are solved by relying on agreed upon business processes, Scrum events, and project rules" (*Res3*).

Technical conflicts between functions were described as being resolved primarily through collaborative technical forums such as problem-solving meetings and design reviews. Respondents noted that bringing different technical perspectives together supports shared understanding and quicker resolution. One participant stated:

"Technical conflicts of functions are generally resolved by working together on problem-solving meetings and design reviews" (*Res4*).

Participants also noted that hybrid work requires additional communication efforts to resolve misunderstandings. Respondents explained that while online tools are used for conflict resolution, follow-up phone calls are sometimes needed to clarify issues more effectively. One respondent stated:

"In hybrid teams, conflicts are resolved by online communication tools, but sometimes follow-up phone calls are required to iron out misunderstandings" (*Res5*).

Agile retrospectives were frequently mentioned as an important mechanism for addressing recurring conflicts. Participants explained that retrospectives allow teams to reflect on repeated issues and implement improvements in future sprints. One respondent noted:

“Agile retrospectives are employed to address and resolve conflicts that repeatedly occur, to guarantee that the lessons are applied into future ‘Sprints’” (*Res6*).

Resource-related conflicts were also reported as requiring leadership intervention. Respondents explained that leaders often resolve such conflicts by prioritizing tasks and reallocating team members or resources to reduce tension. One participant stated:

“Leaders make their intervention in cases where there are resource conflicts, by prioritizing tasks and redeploying team members to alleviate any frictions” (*Res7*).

Timing conflicts, particularly in distributed teams, were described as being managed through scheduling adjustments and the use of asynchronous communication methods. Participants explained that shifting meeting times and using structured updates supports fairness and inclusion for remote members. One respondent noted:

“We handle timing conflicts by adjusting meeting times or implementing asynchronous update procedures that help remote teams” (*Res8*).

Cultural and interpersonal conflicts were described as being addressed through coaching, mentoring, and team-building initiatives. Respondents indicated that these approaches help strengthen relationships and improve understanding across diverse team members. One participant explained:

“Cultural or interpersonal conflicts are handled through coaching and mentoring exercises and in some instances, team building exercises” (*Res9*).

Finally, respondents emphasized that escalation beyond the team level is generally avoided unless necessary. Participants reported that only unresolved disputes are escalated to project managers or senior management. One respondent stated:

“Escalation to project managers or senior management occurs only for disputes that cannot be settled at a team level” (*Res10*).

Table 12 Conflict Management and Resolution Strategies in Hybrid Cross-Functional Teams

Statement	Count
Conflicts are primarily resolved through meetings, dialogue, and joint problem-solving	4
Agile processes and retrospectives help address recurring conflicts	2
Leadership intervention helps manage resource, cultural, and interpersonal conflicts	3
Flexible coordination practices help manage hybrid-specific conflicts	1

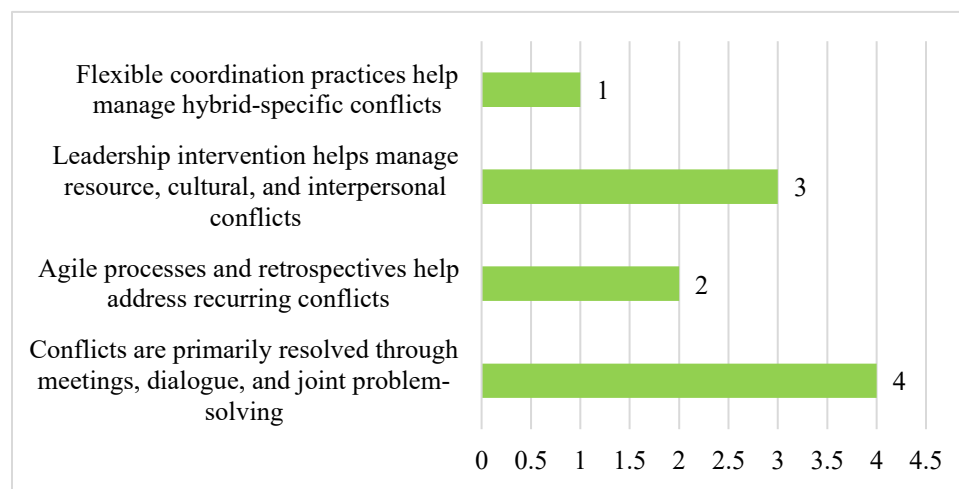


Figure 10 Graph representation of Managerial Practices that Improve Cross-Functional Collaboration and Reduce Silos

4.5 Theme 5: Managerial Practices for Enhancing Collaboration and Adaptability

This theme explores the managerial practices and leadership behaviors that support collaboration, adaptability, and resilience in cross-functional automotive automation projects. The findings provide insight into how managers facilitate cooperation across functions, motivate team members, and enable teams to respond effectively to changes and uncertainties common in technologically complex and hybrid project environments.

The questions under this theme were designed to identify the human-centered managerial practices that help hybrid teams remain aligned and effective, while also gathering recommendations for project and product managers leading cross-functional automotive automation teams. Detailed findings related to these questions are discussed below.

Q10. What practices or behaviors have been most effective in improving collaboration across functions?

Managerial Practices that Strengthen Cross-Functional Collaboration and Adaptability

Respondents described several managerial practices that were perceived as highly effective in strengthening collaboration across functions. Participants emphasized that successful cross-functional cooperation depends on consistent alignment, open communication, clear role definition, shared ownership, transparency, and leadership behaviors that focus on enabling teamwork rather than controlling it. In addition, respondents highlighted the importance of adaptability and continuous improvement in supporting teams operating in dynamic hybrid environments.

A frequently mentioned practice was holding regular cross-functional alignment meetings. Participants explained that such meetings help ensure that all teams share a common understanding of project goals, priorities, and expectations, which strengthens collaboration. One respondent stated:

“Regular cross-functional alignment meetings fostered a common understanding of goals and priorities across all teams, which greatly enhanced collaboration” (*Res1*).

Open communication was also identified as a key managerial behavior. Respondents emphasized that when managers promote psychological safety and encourage questions, team members are more likely to raise issues early, reducing misunderstandings across functions. One participant noted:

“Open communication, besides making people feel safe to ask questions, allowed team members to raise issues at the beginning itself, reducing misunderstandings between functions” (*Res2*).

Several respondents highlighted the importance of defining roles and responsibilities clearly. Participants explained that when responsibilities are well defined and deliverables are shared across functions, silos are minimized and cooperation improves. One respondent stated:

“In addition, defining roles and responsibilities clearly, with shared ownership of deliverables, helped to minimize silos and increase cooperation across teams” (*Res3*).

Joint planning sessions were also frequently mentioned, particularly those involving software, hardware, and quality teams. Participants noted that collaborative planning improves coordination and reduces later-stage conflicts by ensuring that functional expectations are aligned early. One respondent explained:

“Joint planning sessions involving software, hardware, and quality teams allowed much better coordination and reduced conflicts later on in the stage” (*Res4*).

Leadership behaviors that focus on facilitating collaboration rather than monitoring progress were also described as effective. Respondents explained that leaders who remove barriers and support coordination help teams resolve issues more quickly and work more cohesively. One participant noted:

"Leaders who actively facilitated collaboration rather than tracking progress helped teams resolve issues faster and function much more cohesively" (*Res5*).

Agile practices such as sprint reviews and retrospectives were highlighted as important mechanisms for transparency and continuous improvement. Participants noted that these practices support alignment across functions by encouraging teams to reflect on performance and implement improvements collaboratively. One respondent stated:

"Agile practices such as sprint reviews and retrospectives encouraged transparency and continuous improvement across functions" (*Res6*).

Recognition and appreciation of contributions from different functions were also described as improving motivation and collaboration. Respondents explained that when managers acknowledge the value of each function, teams are more willing to support one another. One participant stated:

"The recognition and valuation of contributions from the different functions improved motivation and encouraged teams to support one another" (*Res7*).

Participants further emphasized the value of transparency through shared dashboards and tools. Respondents noted that providing visibility into project progress helps align teams, reduces duplication of effort, and improves coordination. One respondent explained:

Visibility of project progress was given to the management through shared dashboards and tools, thereby aligning the teams and reducing duplication of effort” *(Res8)*.

Cross-functional learning and knowledge sharing were also highlighted as effective managerial practices. Participants explained that encouraging teams to share knowledge improves mutual understanding and strengthens teamwork across disciplines. One respondent noted:

"The managers that encouraged the idea of cross-functional learning and knowledge sharing enhanced mutual understanding and teamwork” *(Res9)*.

Finally, adaptability in leadership was described as essential in dynamic automotive automation environments. Respondents explained that managers who adjust strategies based on team input and evolving project needs help keep teams synchronized and resilient in uncertain conditions. One participant stated:

“Being adaptable in leading and adjusting project strategies in response to input from the teams kept everyone synchronized and in harmony in a dynamic environment” *(Res10)*.

Table 13 Managerial Practices that Improve Cross-Functional Collaboration and Reduce Silos

Statement	Count
Cross-functional meetings, joint planning, and shared dashboards enhance coordination and reduce conflicts	4
Clear roles, shared ownership, and open communication minimize silos and misunderstandings	2
Agile practices, recognition, and knowledge sharing foster collaboration, motivation, and continuous improvement	2
Managerial adaptability helps teams remain synchronized in dynamic project environments	2

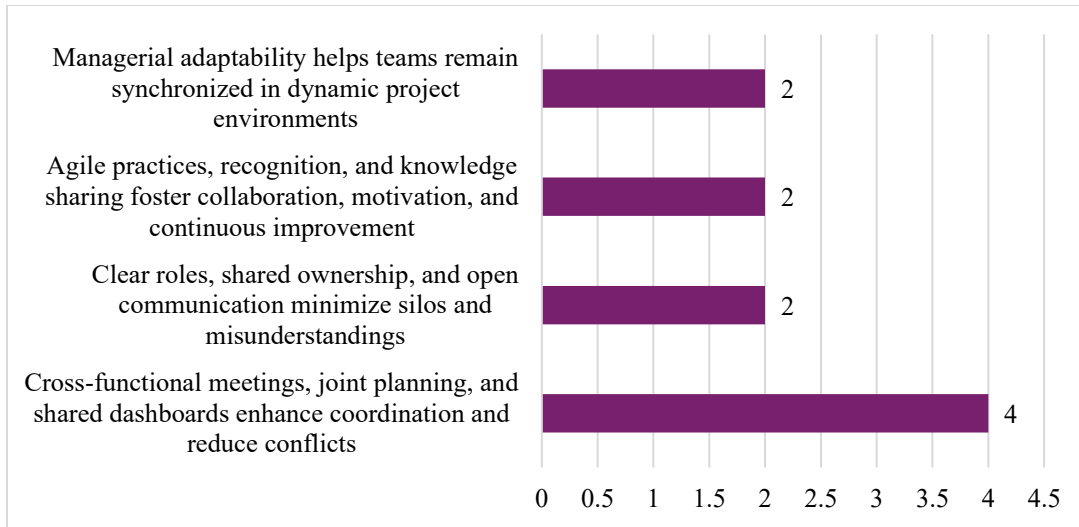


Figure 11 Managerial Practices that Improve Cross-Functional Collaboration and Reduce Silos

Q11. How do teams adapt to changes or uncertainty in automotive automation projects?

Adaptation Strategies for Managing Change and Uncertainty in Automotive Automation Projects

Respondents explained that adaptation in automotive automation projects is largely driven by Agile practices, frequent communication, cross-functional collaboration, and leadership behaviors that promote flexibility and psychological safety. Participants emphasized that automotive automation projects often involve evolving requirements, unexpected technical challenges, and shifting timelines, requiring teams to remain responsive and aligned despite uncertainty. Overall, the findings suggest that teams adapt most effectively when they combine structured Agile planning with open communication, shared understanding, and supportive leadership.

A common adaptation strategy described by respondents was breaking complex changes into smaller tasks and reprioritizing work through sprint planning. Participants explained that this approach helps maintain progress and reduces disruption when new requirements arise. One respondent noted:

“Teams adapt by breaking complex changes into smaller tasks and reprioritizing work during sprint planning to maintain momentum” (*Res1*).

Frequent communication was also emphasized as essential for responding quickly to change. Respondents explained that quick alignment meetings and continuous updates allow teams to adjust efficiently when project requirements or timelines shift. One participant stated:

“Frequent communication and quick alignment meetings enable teams to react effectively when changes occur in project requirements or timelines” (*Res2*).

Cross-functional collaboration was repeatedly highlighted as critical for effective adaptation. Participants noted that when changes occur, teams must jointly assess impacts, review technical implications, and agree on solutions across functions. One respondent explained:

"Cross-functional collaboration is an important part of adaptation because teams review impacts and technical solutions together" (*Res3*).

Several participants described Agile methodologies as central to adaptation, particularly through iterative development and continuous feedback. Respondents indicated that Agile frameworks enable teams to adjust progressively without disrupting the entire project structure. One participant noted:

“Agile methodologies like iterative development and continuous feedback allow teams to adapt easily whenever there are changes in the project requirements” (*Res4*).

Leadership behaviors were also described as supporting adaptability, particularly when leaders encourage experimentation and allow teams to explore alternative approaches under uncertainty. Participants noted that this increases flexibility and improves problem-solving capacity. One respondent stated:

"Leaders enable adaptability by encouraging experimentation and permitting teams to explore different methods in situations of uncertainty" (*Res5*).

Retrospectives were further described as important mechanisms for learning and improvement. Respondents explained that regular reflection enables teams to evaluate what is working well and adjust practices to better handle future uncertainties. One participant explained:

“Regular retrospectives allow the team to look back and evaluate what has been working well and make adjustments to improve handling potential uncertainties that might arise down the road” *(Res6)*.

Clear prioritization from management was also highlighted as essential when teams face change. Respondents emphasized that when priorities are communicated clearly, teams remain focused and avoid confusion during uncertain situations. One participant noted:

“Having clear priorities from the management helps the team concentrate on the most important jobs to avoid confusion when encountering change” *(Res7)*.

Participants also described the importance of shared documentation and transparent communication to support adaptation. Respondents noted that when changes are documented and communicated clearly, all team members understand the implications and can align their work accordingly. One respondent stated:

“Teams rely on shared documentation and transparent communication to ensure everyone understands changes and their implications” *(Res8)*.

Psychological safety was also identified as a key factor supporting adaptation. Participants explained that when team members feel safe to raise concerns early, teams can respond quickly and prevent problems from escalating. One participant stated:

“When there is psychological safety in the team, it encourages team members to voice concerns early, allowing the team to adapt and prevent matters from escalating” *(Res9)*.

Finally, respondents emphasized that strong leadership support and trust contribute to resilience under difficult conditions. Participants explained that when leaders provide consistent support, teams are more capable of persevering through unexpected technical and operational challenges. One respondent noted:

“Strong leadership support and trust enable teams to persevere in tough circumstances, including unexpected technical and operational difficulties” *(Res10)*.

Table 14 Team Adaptation Strategies for Managing Change and Uncertainty in Automotive Automation Projects

Statement	Count
Breaking work into smaller tasks, reprioritizing, using agile methods, and retrospectives enhances workflow flexibility and continuous improvement	3
Frequent communication, shared documentation, alignment meetings, and cross-functional collaboration improve coordination and clarity	3
Leadership support, encouragement of experimentation, and clear priorities enable adaptability and team resilience	3
Psychological safety and open environment encourage raising concerns early, allowing proactive adaptation	1

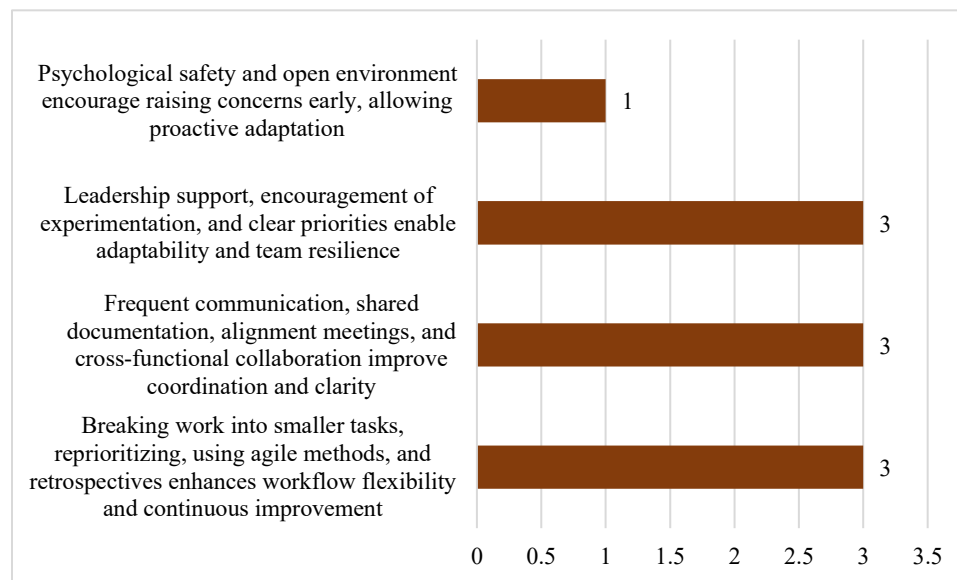


Figure 12 Graph representation of Recommendations for Product and Project Managers Leading Hybrid Automotive Teams

Q12. Based on your experience, what recommendations would you give to product or project managers leading hybrid automotive teams?

Recommendations for Product and Project Managers Leading Hybrid Automotive Automation Teams

Respondents provided several recommendations for product and project managers leading hybrid cross-functional automotive automation teams. Overall, participants emphasized that effective leadership in hybrid environments requires clear communication, trust-building, role clarity,

consistent alignment, psychological safety, Agile adaptability, inclusion of remote team members, proactive conflict management, and attention to team well-being. The findings suggest that human-centered managerial practices are essential for sustaining motivation, collaboration, and project success in complex automotive automation settings.

A frequently emphasized recommendation was the importance of clear and consistent communication across all functions. Participants explained that managers must ensure expectations are understood and reinforced across different teams, particularly in hybrid environments where misunderstandings can persist longer. One respondent stated:

"Product and project managers must communicate and make sure expectations are reinforced in all functions, including in a hybrid space" (*Res1*).

Trust-building was also identified as a critical leadership responsibility. Respondents recommended that managers should establish trust by remaining transparent in decision-making and accessible to team members, even when working remotely. One participant noted:

"Managers need to spend time establishing trust amongst team members by being transparent in decisions and accessible for conversations, even when working remotely" (*Res2*).

Several respondents emphasized the need for clear role definition and ownership from the beginning of the project. Participants explained that establishing responsibilities early reduces confusion and improves collaboration across functions. One respondent stated:

"Defining roles, responsibilities, and ownership clearly from the outset helps ensure there are no confusing moments during team collaboration" (*Res3*).

Participants also highlighted the value of regular cross-functional check-ins. Respondents explained that frequent alignment meetings help maintain shared priorities and allow issues to be identified early, especially in distributed teams. One participant explained:

"Cross-functional check-ins on a regular basis are very important for aligning priorities and for surfacing potential issues especially when teams are distributed" (*Res4*).

Promoting open feedback and psychological safety was another key recommendation. Respondents emphasized that managers should encourage team members to share ideas and concerns freely, which supports collaboration and adaptation. One participant noted:

“Managers must promote open feedback and build a psychological environment where team members feel free to air their views and suggestions” **(Res5)**.

Agile adaptability was also described as essential in managing uncertainty in automotive automation projects. Respondents recommended that managers embrace agile principles and remain flexible in planning to help teams respond effectively to changing requirements and technological uncertainty. One respondent stated:

"By embracing the agile philosophy and being adaptable in planning, teams can respond better to changes and technology uncertainty" **(Res6)**.

Inclusion of remote team members was highlighted as particularly important in hybrid environments. Respondents recommended that leaders ensure equal participation and representation for remote employees in meetings and decision-making processes. One participant explained:

“Leaders must be aware of the difficulties experienced by telecommuters and provide equal representation in meetings and decision-making” **(Res7)**.

Cross-functional learning and knowledge sharing were also emphasized as practices that strengthen teamwork. Respondents noted that enabling knowledge exchange across disciplines improves mutual understanding and supports collaboration in multidisciplinary teams. One respondent stated:

“Enabling the sharing of knowledge and cross-function learning facilitates inter/multi-disciplinary knowledge bridging and improves team working” **(Res8)**.

Proactive conflict and misalignment management was also identified as an important leadership responsibility. Participants recommended that managers address issues early by encouraging constructive dialogue rather than allowing tension to accumulate. One respondent explained:

"Managers must be proactive in dealing with conflict and misalignment by encouraging constructive dialogue, instead of letting things build up" **(Res9)**.

Finally, respondents emphasized the need to balance performance expectations with team well-being. Participants explained that supporting employee wellness is critical for sustaining

motivation, engagement, and long-term project success in hybrid environments. One respondent stated:

“Balancing performance expectations with attention to team well-being is critical for sustaining motivation and long-term project success in hybrid teams” (*Res10*).

Table 15 Recommendations for Product and Project Managers Leading Hybrid Automotive Teams

Statement	Count
Communicate clearly and transparently across all functions.	2
Define roles and responsibilities clearly from the start.	1
Hold regular cross-functional check-ins to keep teams aligned.	2
Encourage open feedback and create psychological safety for all team members.	2
Use Agile methods and stay flexible when changes occur.	1
Promote knowledge sharing and cross-functional learning.	1
Balance performance goals with team well-being.	1

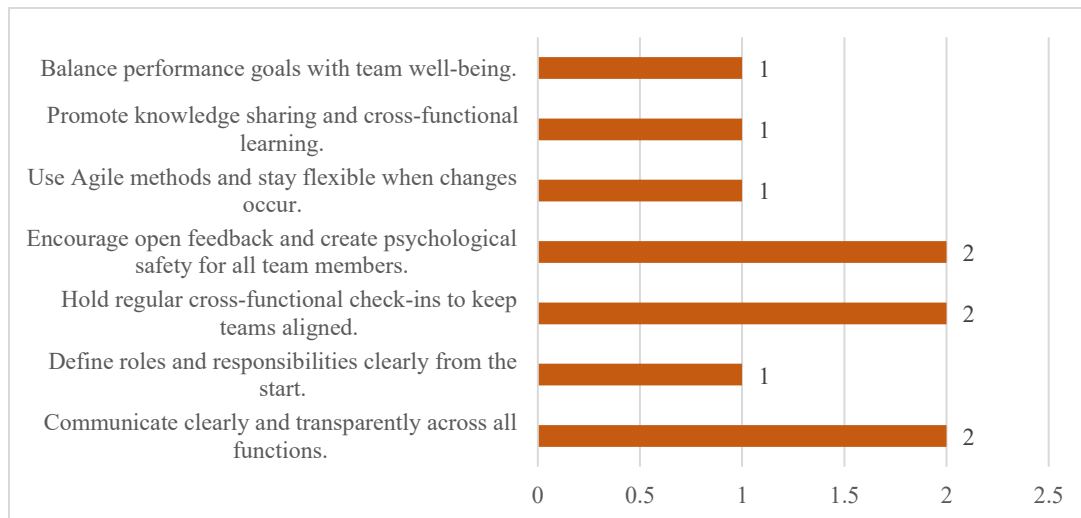


Figure 13 Graph representation of Recommendations for Product and Project Managers Leading Hybrid Automotive Teams.

4.6 Summary

Chapter 4 presents the qualitative findings derived from interview data exploring leadership, communication, coordination, conflict management, and managerial practices in cross-functional

automotive automation projects operating under hybrid or distributed working arrangements. The results were structured into five major themes, each supported by direct participant quotations and summarized through frequency-based tables. Under Theme 1, findings revealed that leadership approaches were largely collaborative, inclusive, and adaptive, with respondents describing leaders as accessible and supportive while also maintaining structure and goal orientation. Leadership behaviors were consistently linked to trust-building, motivation, and improved collaboration, particularly when leaders enabled cross-functional participation, recognized contributions, and removed barriers rather than relying on micromanagement. However, the findings also highlighted that leadership effectiveness increased when technical understanding and decision clarity were present, especially in complex system integration contexts where ambiguity could otherwise create delays or reduce accountability.

Themes 2 and 3 highlighted that communication and coordination in hybrid environments rely heavily on formal mechanisms such as sprint meetings, stand-ups, milestone reviews, dashboards, and digital platforms like Teams and Jira. While these tools supported information flow and transparency, participants reported significant challenges including time zone delays, slower response times, reduced informal interactions, and weaker relationship-building. These barriers often contributed to misinterpretations, fragmented information, and difficulties in addressing urgent issues quickly. Theme 3 further emphasized that role alignment and responsibility clarity are critical in cross-functional projects, as misalignment frequently caused integration failures, duplicated work, rework, and delays in approvals. Theme 4 demonstrated that conflicts commonly emerge from task priorities, process differences, technical misunderstandings, decision authority ambiguity, and cultural communication differences. Teams typically resolved conflicts through weekly coordination meetings, joint problem-solving, agile retrospectives, and escalation only when necessary. Finally, Theme 5 showed that collaboration and adaptability were strengthened through cross-functional alignment meetings, clear ownership structures, shared dashboards, open feedback cultures, agile practices, and psychological safety. Respondents strongly recommended that project and product managers focus on transparency, trust-building, role clarity, equal inclusion of remote members, proactive conflict handling, and balancing performance demands with team well-being to sustain long-term hybrid team success.

CHAPTER-5

5 Discussion and Conclusion

5.1 Discussion

The findings of this study provide empirical support for the analytical framework developed in Chapter 2, demonstrating that human-centred leadership and communication practices significantly influence collaboration in hybrid automotive automation project teams through their impact on team dynamics. The results indicate that leadership and communication do not directly generate collaboration; rather, they shape the psychological and relational conditions such as trust,

motivation, psychological safety, and conflict management through which collaboration becomes possible. In hybrid project environments characterized by technological complexity, cross-functional interdependence, and reduced face-to-face interaction, these human-centred managerial behaviours function as critical stabilizing mechanisms that compensate for structural fragmentation and communication constraints inherent in distributed work arrangements.

The analysis revealed that effective leadership in hybrid automotive automation projects extends beyond traditional task-oriented control and performance monitoring. Participants consistently emphasized accessibility, transparency, empathy, and supportive engagement as defining characteristics of impactful leadership. This aligns with contemporary leadership literature, which highlights a shift from directive and control-based project management toward relational and empowerment-oriented leadership approaches. However, the present findings add contextual depth by demonstrating that in hybrid settings, leadership behaviours carry amplified significance due to reduced informal interaction. In co-located teams, trust and cohesion often develop organically through daily social contact. In hybrid environments, such spontaneous relational development is limited, and leadership behaviours become primary signals shaping perceptions of fairness, reliability, and inclusion. Leaders who maintained regular engagement, clarified expectations, and encouraged open dialogue were more likely to foster psychological safety and motivation among cross-functional team members.

The findings further suggest that empowerment and autonomy are particularly important in hybrid automotive projects, where teams must adapt to rapidly evolving technological requirements and interdependent workflows. Leaders who provided strategic clarity while allowing discretion in execution enhanced intrinsic motivation and collective ownership. Conversely, overly directive leadership styles were associated with reduced initiative and diminished engagement. This supports existing theoretical perspectives on transformational and servant leadership while reinforcing the proposition that human-centred leadership positively influences team dynamics in hybrid project contexts.

Communication practices emerged as equally central to collaborative effectiveness. Participants described communication not only as a mechanism for transmitting information but also as a relational process shaping alignment and trust. Hybrid work arrangements increase reliance on digital platforms, yet the findings indicate that technology alone does not ensure coordination.

Where communication lacked clarity, consistency, or transparency, teams experienced duplication of effort, misunderstandings, and delays in decision-making. Structured communication routines such as regular cross-functional alignment meetings and clearly defined information channels were perceived as essential in maintaining shared understanding. These findings reinforce established communication theories emphasizing the importance of feedback loops and clarity in reducing ambiguity, particularly in environments with limited non-verbal cues.

Importantly, inclusive communication practices played a critical role in hybrid team dynamics. Participants noted that remote members could feel less visible or less involved in decision-making processes if discussions were dominated by those physically present. Leaders who actively invited contributions from all participants and ensured equitable information access mitigated these risks. This finding extends existing hybrid work literature by demonstrating that inclusiveness is not only an ethical imperative but also a functional requirement for sustaining engagement and collaboration across distributed teams.

The mediating role of team dynamics was strongly supported by the empirical data. Trust emerged as a foundational element enabling effective cross-functional collaboration. Both cognitive trusts, grounded in perceptions of competence, and affective trust, based on interpersonal relationships, were identified as critical. In hybrid automotive projects, trust required deliberate reinforcement through consistent communication and ethical leadership behaviours. Psychological safety was similarly central, particularly in complex automation projects where rapid iteration and problem detection are common. Teams characterized by high psychological safety were more willing to raise technical concerns early, share uncertainties, and collaboratively solve problems. In contrast, environments lacking such safety experienced prolonged issue resolution and reduced adaptive capacity.

Conflict dynamics further illustrated the mediating influence of team processes. Task-related disagreements were often constructive when managed through open dialogue and shared problem-solving. However, relational tensions intensified when communication breakdowns persisted or when digital interactions were misinterpreted. Effective leaders reframed conflicts around shared project objectives, thereby preventing escalation and preserving cohesion. These observations confirm that leadership and communication practices shape the internal relational climate of teams, which in turn determines the quality of collaboration.

Collaboration itself emerged as an outcome characterized by coordinated action, shared ownership, adaptive decision-making, and collective problem-solving across engineering, quality, operations, and product management functions. Rather than being a static attribute, collaboration appeared as an emergent property resulting from aligned leadership, structured communication, and positive team dynamics. Projects where human-centred practices were consistently applied demonstrated greater resilience in responding to shifting requirements and technological uncertainty. This finding reinforces the proposition that positive team dynamics enhance collaboration and underscores the strategic importance of relational competencies in hybrid project environments.

From a theoretical perspective, this study contributes to project management research by empirically validating an integrated framework linking human-centred leadership, communication practices, team dynamics, and collaboration within hybrid automotive automation contexts. While prior literature has examined these constructs individually, the present findings demonstrate their interdependence and highlight the mediating role of team dynamics. Moreover, the study reframes leadership in hybrid environments as a relational stabilizer that compensates for reduced informal interaction and fragmented communication structures.

Managerially, the findings suggest that project managers in hybrid automotive automation environments should prioritize trust-building behaviours, establish clear and inclusive communication routines, balance autonomy with strategic alignment, and proactively address relational tensions. Technical competence and methodological rigor remain important, but they are insufficient without deliberate cultivation of the human dimension of project management.

Several limitations should be acknowledged. The qualitative design and purposive sampling strategy limit generalizability beyond similar automotive automation contexts. Additionally, the findings reflect participant perceptions and may be influenced by organizational culture or role-specific perspectives. Future research could extend this work through longitudinal studies examining trust development over time or through quantitative testing of the proposed framework across industries.

Overall, the discussion confirms that collaboration in hybrid automotive automation projects is fundamentally shaped by human-centred leadership and communication practices. In technologically intensive and distributed work environments, the social and relational dimensions of project management emerge as decisive factors in sustaining trust, adaptability, and

performance. These findings reinforce the central argument of this thesis: that effective collaboration in contemporary hybrid project teams depends not only on technical systems and processes but, critically, on the quality of leadership and communication that shape team dynamics.

5.2 Conclusion

The present study set out to explore how human-centered leadership and communication practices influence team dynamics and collaboration within hybrid automotive automation project teams. Drawing on a qualitative research design grounded in semi-structured interviews with experienced professionals across product management, engineering, quality assurance, and operations, the research sought to integrate theoretical insights with lived managerial experiences. The findings provide a coherent and empirically supported response to the research problem and questions, while also contributing to contemporary debates on hybrid work, project leadership, and cross-functional collaboration.

At its core, this thesis demonstrates that collaboration in hybrid project environments does not emerge automatically from technical structures, digital tools, or formal governance mechanisms. Instead, collaboration is a relational outcome shaped by leadership behaviors and communication practices that influence internal team dynamics. The empirical results confirm the analytical framework developed in Part 1, particularly the mediating role of team dynamics between human-centered leadership and communication on one side, and collaborative effectiveness on the other. Leadership and communication practices were shown to affect trust, psychological safety, motivation, and conflict management processes, which in turn determined the extent to which cross-functional teams could coordinate effectively, adapt to uncertainty, and collectively solve problems.

One of the central conclusions of this study is that hybrid project environments amplify the importance of relational competence in leadership. Reduced face-to-face interaction limits informal trust-building and spontaneous alignment. As a result, leadership behaviors become more visible and more consequential. Participants consistently emphasized the value of leader accessibility, transparency, ethical conduct, and empowerment. These behaviors were associated with stronger trust and engagement, particularly in distributed teams where visibility of effort and

intention may otherwise be limited. The findings therefore suggest that in hybrid automotive automation projects, leadership operates less as a supervisory mechanism and more as a relational stabilizer that compensates for structural fragmentation.

Similarly, communication practices were found to perform both instrumental and relational functions. While formal communication mechanisms ensured coordination of tasks and dependencies across engineering and operational functions, relational aspects of communication such as tone, inclusiveness, responsiveness, and clarity shaped psychological safety and mutual understanding. Hybrid settings, characterized by reliance on digital communication platforms, increased the risk of misinterpretation and information overload. The study demonstrates that deliberate structuring of communication routines, transparent information sharing, and inclusive dialogue are essential for sustaining alignment and minimizing conflict. In this sense, communication in hybrid project teams is not merely a channel for transmitting information but a mechanism for constructing shared meaning and reinforcing collective responsibility.

The research further confirms that team dynamics constitute the central mechanism through which leadership and communication translate into collaborative outcomes. Trust both cognitive and affective emerged as foundational to effective cross-functional interaction. Psychological safety was particularly critical in technology-intensive automotive automation projects, where rapid problem detection, iterative development, and innovation require team members to speak openly about uncertainties and risks. Where leaders modeled openness and encouraged dialogue, teams demonstrated greater adaptability and resilience. Conversely, where psychological safety was weak, conflicts persisted longer, and collaboration deteriorated.

Conflict dynamics provided additional insight into the complexity of hybrid teamwork. Task-related disagreements were often productive when managed constructively, contributing to improved decision-making and technical rigor. However, relational conflicts intensified in contexts of unclear communication or misaligned expectations. Effective conflict management required both structured dialogue and leader intervention focused on shared goals rather than individual positions. This underscores the interdependent nature of leadership, communication, and team processes in determining collaborative quality.

Beyond validating the proposed analytical framework, this study contributes to theory by offering an integrated perspective that connects human-centered leadership, communication practices, team dynamics, and collaboration within hybrid project contexts. While existing literature has examined these constructs independently, this research demonstrates their systemic interaction in technologically complex environments. The findings also extend hybrid work literature by highlighting the heightened strategic importance of relational leadership in distributed project settings, particularly in industries characterized by high task interdependence and rapid technological change.

From a managerial standpoint, the conclusions of this study emphasize that effective hybrid project management requires intentional cultivation of the human dimension of teamwork. Project managers must move beyond a narrow focus on technical coordination and adopt practices that actively build trust, clarify expectations, encourage inclusive participation, and address relational tensions early. Structured communication routines, clear role definitions, and empowerment balanced with accountability were identified as practical mechanisms for enhancing collaboration. Organizations operating in hybrid automotive automation environments should therefore invest in leadership development programs that strengthen emotional intelligence, inclusive communication skills, and conflict facilitation competencies.

It is important to acknowledge the limitations of this research. The qualitative design, while enabling rich contextual insight, limits statistical generalization. The purposive sampling strategy focused on experienced professionals within automotive automation contexts, and findings may differ in other industries or cultural environments. Additionally, the study captures perceptions at a particular point in time; longitudinal research could further illuminate how trust and collaboration evolve across project lifecycles. Nevertheless, the depth and consistency of the data provide a robust foundation for conceptual insights and practical implications.

Future research may build upon this work by quantitatively testing the proposed framework across diverse sectors, comparing hybrid and fully remote project teams, or examining the role of digital collaboration tools in shaping psychological safety. Comparative cross-cultural studies may also deepen understanding of how leadership and communication practices operate in different

institutional settings. Such research would further refine the understanding of human-centered management in contemporary project environments.

In conclusion, this thesis affirms that the success of hybrid automotive automation projects depends not solely on technological infrastructure or methodological rigor, but fundamentally on the quality of human interaction shaped by leadership and communication practices. In environments marked by distributed work arrangements, complex interdependencies, and rapid technological evolution, collaboration emerges as a relational achievement rather than a procedural outcome. Human-centered leadership and communication are therefore not supplementary soft skills, but strategic capabilities essential to sustaining trust, adaptability, and performance in hybrid project teams. Through its integrated analytical framework and qualitative empirical insights, this study contributes to a more nuanced understanding of project management in the digital age and reinforces the enduring importance of the human dimension in organizational effectiveness.

6 References

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